Supporting information for *The Selling closure: a complete operator set for reindexing, and the price of pseudo-symmetry*

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All code is in `agentsg` (Zwart, 2026), version 0.5.0, `github.com/phzwart/agentsg`; the scripts named below are in its directory `manuscript/supplement/benchmark`, and the checks in the package `sellingproofs` (Section S7). Table, section and equation numbers of the main text are cited as there. Conorm vectors are written in the slot order da, db, dc, ab, ac, bc of the main text, and strains in per cent. Section S7 lists, for every section, table and figure, the check that reproduces its numbers. Terms are those of the main text (Table 2): a *step* is a Selling step, and at a zero conorm a boundary transformation; a *face* is the set of conorm vectors with a given zero pattern, a boundary of the cone of reduced conorms; a *difference-wall member* is a member of a widened set that is reachable from the reduced cell only through a non-member, across the wall where two small conorms are equal.

S1. Catalogues of shapes and specialisations

Cell shapes. Table S1 lists the shapes of the ten sorts with more than one, each as the cell with the most right angles and no acute angle. The three shapes of a primitive monoclinic lattice are the three cell choices of *International Tables* (Hahn,

2006, Section 2.2.16); the extra shapes of primitive orthorhombic, tetragonal and cubic lattices are monoclinic-looking cells with two right angles; C-centred orthorhombic has two primitive shapes; the boundary sorts of body-centred orthorhombic, C-centred monoclinic and triclinic have two or three cells with a single right angle each, for triclinic the known instability of the reduced cell when an angle is near 90° (Andrews *et al.*, 1980).

Table S1. *The cell shapes of the sorts with more than one, for the cells named; each shape is given as the cell with the most right angles and all angles* $\geq 90°$*, shortest edges first when several qualify and b unique for the monoclinic-looking ones, with the number n of members (up to relabellings) that have it. Centred cells are given by their primitive cells; the boundary sorts are specified by their conorms in* $Å^2$.

sort	lattice	n	shape
cP	30 Å	4	30, 30, 30; 90, 90, 90
		12	30, 30, 42.4; 90, 135, 90
tP	30, 30, 50 Å	4	30, 30, 50; 90, 90, 90
		4	30, 30, 58.3; 90, 121.0, 90
		8	30, 50, 42.4; 90, 135, 90
oP	30, 40, 55 Å	4	30, 40, 55; 90, 90, 90
		4	30, 40, 62.6; 90, 118.6, 90
		4	30, 55, 50; 90, 126.9, 90
		4	40, 55, 50; 90, 143.1, 90
oS	C 30, 40, 50 Å	2	25, 50, 25; 90, 106.3, 90
		4	25, 50, 30; 90, 126.9, 90
mP	30, 40, 50 Å, 100°	2	30, 40, 50; 90, 100, 90
		2	30, 40, 53.7; 90, 113.4, 90
		2	50, 40, 53.7; 90, 146.6, 90
oI (O3)	900, 1600, 0, 0, 1600, 900	1	50, 50, 50; 111.1, 129.8, 90
		2	42.4, 50, 50; 90, 115.1, 115.1
mC (M2B)	900, 1600, 2500, 0, 1600, 900	1	50, 50, 70.7; 104.7, 116.9, 90
		1	50, 50, 65.6; 105.9, 105.9, 90
mC (M3)	2500, 1600, 0, 0, 1600, 900	1	50, 50, 64.0; 120, 90, 111.1
		2	50, 56.6, 58.3; 90, 108.0, 124.4
aP (A2)	2500, 400, 1200, 0, 1600, 900	1	36.1, 60.8, 64.0; 114.3, 90, 114.2
		1	36.1, 56.6, 64.0; 116.2, 90, 101.3
aP (A3)	2500, 400, 0, 0, 1600, 900	1	36.1, 50, 53.9; 90, 101.9, 119.9
		1	44.7, 50, 53.9; 90, 99.6, 135.7
		1	36.1, 44.7, 58.3; 90, 115.3, 104.4

Specialisations. The conorms of any member, in any labelling, are integer combinations of the reduced cell's conorms, hence linear in the free parameters of the sort (the letters of Table S10). Imposing another sort's zeros and equalities on one member is a homogeneous linear system in those parameters, and a solution with non-negative parameters, a non-singular lattice and a resulting lattice of the target sort and no

higher is a specialisation. Table S2 lists the conditions of codimension one, on 36 pairs of sorts. Each is a parameter vanishing or two parameters becoming equal. The classical values appear: body-centred tetragonal is face-centred cubic at $c/a = \sqrt{2}$ and body-centred cubic at $c/a = 1$; C-centred orthorhombic is hexagonal at $b = \sqrt{3}\,a$ and tetragonal at $b = a$; rhombohedral is cubic at $\alpha = 60°$, $90°$ and $109.47°$. The monoclinic conditions come in threes, one per shape: with unique axis c and plane lattice a, b, the lattice is C-centred orthorhombic when $|a| = |b|$, $|a| = |a + b|$ or $|b| = |a + b|$, and primitive orthorhombic when $a \perp b$, $a \perp (a+b)$ or $b \perp (a+b)$ — the textbook condition $a = c$ read in each of the three cell choices. The table also shows which sorts are one condition from a much higher symmetry: C-centred monoclinic from tetragonal, body-centred tetragonal and rhombohedral; the body-centred orthorhombic boundary sort from face-centred cubic; triclinic on a wall from C-centred monoclinic.

Table S2. *All specialisations of codimension one between the 24 sorts. Conditions are on the letters of the sort's pattern in Table S10; "shapes" is the number of distinct cell shapes of the starting sort through which the conditions are reached, and the last column reads the condition in the cell of the conventional description where that is short.*

from	to	conditions	shapes	reading
aP A1	aP A2	any letter $= 0$	1	a right angle appears
aP A2	aP A3	$v = 0$	1	a second right angle
aP A2	mC M2A	$s = t$, $r = u$	1	
aP A2	mP M4	r, s, t or $u = 0$	1	
aP A3	mC M3	any two letters equal	3	one per shape
mP M4	oP O5	$r = 0$, $s = 0$, $t = 0$	3	$a \perp b$, $a \perp (a+b)$, $b \perp (a+b)$
mP M4	oS O4	$s = t$, $r = t$, $r = s$	3	$\lvert a\rvert = \lvert b\rvert$, $\lvert a\rvert = \lvert a+b\rvert$, $\lvert b\rvert = \lvert a+b\rvert$
mC M1A	oF O1A	$r = t$	1	
mC M1A	oI O2	$u = 0$	1	
mC M1A	oS O4	$r = 0$, $t = 0$	1	
mC M1B	mC M2B	$t = 0$, $u = 0$	1	boundary sort
mC M1B	mC M3	$r = 0$, $s = 0$	1	boundary sort
mC M1B	oF O1A	$r = s$	1	
mC M1B	oI O1B	$t = u$	1	
mC M2A	oI O2	$r = t$	1	
mC M2B	oI O3	$t = 0$	1	
mC M2B	tI T2	$r = s$	1	one step to tI
mC M2B	tP T5	$r = 0$, $s = 0$	1	one step to tP
mC M3	oI O3	$r = t$	1	
mC M3	hR R3	$r = s$, $s = t$	1	one step to hR
oP O5	tP T5	$r = s$, $r = t$, $s = t$	1	$a = b$, $a = c$, $b = c$
oS O4	tP T5	$r = 0$	1	$b = a$
oS O4	hP H4	$r = s$	1	$b = \sqrt{3}\,a$
oI O1B	oI O3	r, s or $t = 0$	1	boundary sort
oI O1B	tI T1	$r = s$, $r = t$, $s = t$	1	two edges equal
oI O2	tI T2	$r = s$	1	
oI O3	cF C3	$r = s$	1	one step to cF
oF O1A	tI T1	$s = t$	1	
oF O1A	tI T2	$s = 0$, $t = 0$	1	
tP T5	cP C5	$r = s$	1	$c = a$
tI T1	cI C1	$r = s$	1	$c/a = 1$
tI T1	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from below
tI T2	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from above
hR R1	cP C5	$r = 0$	1	$\alpha = 90^\circ$
hR R1	cI C1	$r = s$	1	$\alpha = 109.47^\circ$
hR R3	cF C3	$r = s$	1	$\alpha = 60^\circ$

S2. The benchmark, the comparisons and the walks

The benchmark. The directory `manuscript/supplement/benchmark` of the repository holds 780 measured cells with exact references and true operators, in five families: 75 cells measured in a different shape of the same lattice; 660 lattices placed 0.5 to 8 times the error from a condition of Table S2; 20 cells whose only reduced super-

base is a difference-wall member; 16 needles and plates with angular errors; 9 cells of the boundary sorts. Each case carries its Niggli cell. The errors are isotropic in the metric tensor, of median strain 0.1%. A case counts as found when the true proper operator itself is among the candidates. `evaluate_strain.py` scores the closure and the $\{-1, 0, 1\}$ search at the band and at three times each case's error (`results_strain.txt`); `evaluate.py --niggli` tallies the Niggli cells by family and sort (`results_niggli.txt`). The measured Niggli cell is a different cell from the reference's in 45 cases (O1B 23, O3 12, M1B 6, one each at M2B, M3, R3 and T2; 14 of them a change between type I and type II), and the same cell in another basis in 125 (O1B 25, M1A 15, O1A 15, T1 15, M1B 13, O4 9, O2 7, R1 7, M2A 6, T2 6, O3 5, A2 1, A3 1).

The comparisons. `compare_methods.py` runs the searches of Table S14 on the benchmark (`results_compare.txt`) and, with `--stress`, on the stress family (`results_stress.txt`): 60 references drawn from the benchmark, each with independent traceless errors on the reference and the new cell, scaled so that the true operator's price is t times the band, $t = 0.5$, 0.9, 1.1 and 1.5, and a random proper reindexing with entries in $\{-2, \ldots, 2\}$. Each method's candidates are made proper, priced by the geodesic distance after the best scale, and kept within the band; at $t = 1.1$ and 1.5 no method found the true operator, as by construction none should. `window_failures.py` builds the window family (`results_window_failures.txt`): 300 lattices of the sorts M4, O4, H4, O5, T5 and C5 with random parameters, each measured twice with an error of 0.1 to 2%, the new cell in a random proper basis and the reference given as its Niggli cell, classified by the Niggli type of the reference and by the kind of face (with an exact reference in the labelling of the sort's pattern the zeros never sit on d and no window fails); the same 300 lattices, drawn again, with the reference in a random proper basis; and 60 lattices on random two-

and three-zero faces whose closure has a member needing an entry of 3, measured in that member. `coset_rank.py` groups the closure's candidates by the exact symmetries of the underlying lattice (`results_coset.txt`). `niggli_types.py` reduces near-monoclinic cells from their Niggli cells and records where the near-zero conorms land (`results_niggli_types.txt`). `face_labellings.py` gives, for every face, the largest entry of the operators out of the reduced cell as reduction returns it and in its best labelling (`results_face_labellings.txt`). `lookup_benchmark.py` runs the lookup of $\mathcal{U}$ on the benchmark and the stress family and counts the references whose r_1 does not certify it (`results_lookup.txt`). `benchmark_misses.py` lists the benchmark cases that the $\{-1, 0, 1\}$ window and the pipeline miss (`results_benchmark_misses.txt`). `niggli_presentation.py` reduces measured near-oP cells from their Niggli form and from random bases and records where the near-zero conorms land (`results_niggli_presentation.txt`). `lookup_plate_case.py` builds the plate case in which an uncertified lookup misses an operator (`results_lookup_plate_case.txt`). `export_universal_operators.py` writes $\mathcal{U}$ as column operators with their determinants (`universal_operators_columns.json`). `adjacency_radii.py` computes r_1 for the benchmark references and for 2000 PDB cells drawn at random, and times the widening and the reindexing (`results_adjacency_radii.txt`).

The PDB. `pdb_nonisomorphism.py` measures the price between entries of one crystal form (`results_pdb_nonisomorphism.txt`). `pdb_lepage.py` runs the cctbx lattice-symmetry analysis at 3° on every entry and prices the pseudo-symmetries (`pdb_lepage.jsonl`, `results_pdb_lepage.txt`); `pdb_pricing_summary.py` gives the transitions and the fractions within the non-isomorphism band (`results_pdb_pricing.txt`). `pdb_band_controls.py` varies the cut between crystal forms, drops identical cells, weights deposition blocks equally, splits the band by Bravais type, and counts on 3000 same-form pairs how often a wrong class is

cheaper than the identity's (`results_pdb_band_controls.txt`). `pdb_lepage_null.py` compares the Le Page rate of real cells with that of cells whose parameters are shuffled within a space group (`results_pdb_lepage_null.txt`). Table S8 lists the prices by the Delone sort of the Le Page lattice and Table S9 by transition from the reported type.

Walks. Table S3 compares four walks on the measured metric, with a band τ on the conorms, and the construction of Section S4 against brute force.

Table S3. *Four walks on the measured metric, and the construction of Section S4, against brute force on 80 cells with zero conorms at awkward positions (orthorhombic, cubic, plate, monoclinic and hexagonal zero patterns), with a band τ on the conorms at* 0.1 *and* 0.5 *of* $\lambda_1/32$.

procedure	cells with a wrong set
step wherever $\lvert p\rvert \le \tau$, keep only cells within the band	13
step wherever $\lvert p\rvert \le \tau$, filter at the end	13
step wherever $\lvert p\rvert \le 2\tau$, keep only cells within the band	13
step wherever $\lvert p\rvert \le 2\tau$, filter at the end	2
idealise at 4τ, enumerate the exact closure, filter	0

S3. The operators alone

The closure can be described without any lattice. An operator $\mathbf{W}$, written as the new basis in the coordinates of the reduced cell, acts on the six conorms by a fixed integer 6×6 matrix $\mathbf{A_W}$, since every conorm of $\mathbf{W}F_0$ is an integer combination of the conorms of F_0. The reduced cells are the points of the orthant $\{p \ge 0\}$ in conorm space. Selling (1874) showed that every lattice has a conorm vector in this orthant, unique up to the 48 relabellings except on the faces (Delaunay, 1933; Conway & Sloane, 1992), and the closure of a lattice is the set of points of its orbit that lie in the orthant: one in the interior, several on the faces. So $\mathbf{W}$ is an operator of the closure of every lattice on the face of Z exactly when $\mathbf{A_W}$ maps the face $\{p \ge 0,\ p_Z = 0\}$ into the orthant, a sign condition on the entries of $\mathbf{A_W}$ and nothing else.

Enumerating integer matrices with entries in $[-2, 2]$ under that condition, over

all 38 faces, and closing the result under relabelling at both ends, gives a set $\mathcal{U}$ of $6096 = 48 \times 127$ integer matrices with entries in $\{-3, \ldots, 3\}$; per face there are 48×1, 2, 3, 6 and 16 matrices for no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector and three zeros. Both placements of three zeros, the triangle and the path, were enumerated; they describe the same lattices read from an orthogonal cell or from a cell with a face diagonal, each has 16 members, and no relabelling carries one placement into the other. $\mathcal{U}$ is closed under inversion, and every element has a labelling whose basis vectors have entries in $[-2, 2]$. That $\mathcal{U}$ contains every operator between two reduced cells of one lattice, whatever the lattice, does not rest on that window: Lemma A2 of the main text confines the members of every closure to superbase coordinates $s + 2k$ with $|k_i| \leq 3$ relative to a reduced cell, and the enumeration of that whole window, over all 38 faces, finds only members that lie in $\mathcal{U}$ in each of their 48 labellings. In the basis of the reduced cell their entries are in $\{-1, 0, 1\}$ when the zero conorms avoid d, and up to 3 otherwise (Table 4 of the main text). Up to relabelling at both ends $\mathcal{U}$ has nine kinds of element, of step length 0 to 4 (Table S4).

Table S4. *The nine kinds of operator between Selling-reduced cells, up to relabelling at both ends; "matrices" counts the integer matrices of the kind in $\mathcal{U}$, "steps" its length as a word of steps, and the last column the zero patterns in whose closures it occurs.*

kind	matrices	steps	occurs with
1	48	0	every pattern (the relabellings)
2	288	1	every pattern with a zero
3	1152	2	two zeros sharing a vector; three zeros
4	1152	3	three zeros
5	1152	3	three zeros
6	576	3	two zeros sharing a vector; three zeros
7	576	4	three zeros
8	576	4	three zeros
9	576	4	three zeros

Why four. For a lattice with $a \perp b \perp c$ the sixteen members are four orthogonal superbases, differing by the sign of one axis, and twelve with a face diagonal. A step from an orthogonal cell always gives a cell with a face diagonal, so changing the sign of one axis takes three steps, out, across and in, and the diagonal cells attached only to

the far orthogonal cell take one more: from (a, b, c) the members lie at step distances $0, 1, 2, 3, 4$ in numbers $1, 3, 6, 3, 3$, and four is the diameter of the largest face.

S4. A band on the conorms

The simplest widening of the closure is an absolute band τ in Å^2 on the conorms,

$$\mathcal{S}_\tau(\mathbf{G}) = \{\, F \;:\; \text{all six conorms of } F \text{ under } \mathbf{G} \text{ are } \geq -\tau \,\}, \tag{1}$$

Selling reduction with its boundary moved from 0 to $-\tau$. This is an L^∞ band on the conorms, a different norm from the strain of the main text; the two are comparable only for cells whose edges are similar in length. It can be computed by idealising: set every conorm of the reduced superbase F_0 that is at most 4τ to zero, enumerate the exact closure of the idealised lattice Λ^*, and keep the members whose conorms under the measured metric are all at least $-\tau$. The construction is exact when λ_1, the squared length of the shortest lattice vector, exceeds 32τ (Theorem S4.5), a ratio independent of the scale of the cell: for an isotropic cell it allows an angular error of about 1.8°. It fails for anisotropic cells, where the band must cover the conorms between the long vectors: on the 16 needles and plates of the benchmark it declined 8. On lattices of five zero patterns with zero conorms placed at random it was exact at every band from $\lambda_1/32$ to $\lambda_1/6$ in which the idealised lattice existed; on dense two-zero grids idealising at 2τ was already exact, at τ not. For the worked cell at $\tau = 20\,$Å^2, within the hypothesis, it returns the six members of the ring. A new cell constructed so that its only reduced superbase is a difference-wall member of an orthorhombic corner was matched against two stored sets of the reference: the complete set offers three candidates and their negatives; a set built by idealising only at the band has lost the third, $\pm\mathbf{I}$, the true operator.

Bands per slot. A band that scales with the vectors of each superbase, $p_{ij}(F) \geq -\sigma|u_i||u_j|$ on every slot of F (all pairwise angles at least $90° - \arcsin\sigma$), is the natural

angular form of the L^{∞} band, but the same construction applied with each slot of F_0 idealised against its own band is not exact. On the needle $5.58 \times 5.21 \times 131.97\,\text{Å}$, $\gamma = 91.12°$, at $\lambda_1/(\sigma l_a l_c) = 33$ and 40, it returns 6 superbases where brute force finds 12: the conorm ab survives idealisation because its own band is small, while the members built on $a + c$ carry the same dot product on a slot whose band is about 25 times larger. Idealising until the zeroed slots are consistent across the members of the closure repairs this, and on 240 random needles, plates, isotropic and monoclinic cells it is never wrong (exact in 124, refused in 116), while the single-pass construction is wrong in 9 (z57).

For a superbase F and two metrics, $\kappa_F(\mathbf{G}, \mathbf{G}')$ is the largest change among F's six conorms.

Proposition S4.1 (characterisation superbase by superbase). $\mathcal{S}_\tau(\mathbf{G}) = \{F : F \in \mathcal{C}(\mathbf{G}')$ for some $\mathbf{G}'$ with $\kappa_F(\mathbf{G}, \mathbf{G}') \le \tau\}$. *Proof.* ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\kappa_F \le \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) The metric whose conorms in the basis of F are $\max(p_{ij}, 0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite and F is obtuse for it. □

Lemma S4.2 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\le \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \ge 4m^*$ in some class, the excess taken for Λ^*. *Proof.* In the basis of F_0, Lemma A1 of the main text gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, F is obtuse for Λ^*; otherwise some $\gamma_{ij} \ge 1$ multiplies a $p^0_{ij} \ge m^*$. □

Lemma S4.3 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \le 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \le 8\tau$ has a partial sum of squared length at

most $32\tau < \lambda_1$. Lemma A1 in the basis of F gives $0 \le e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \le \tau \sum_N \gamma$. If a large γ were positive, Lemma S4.4(ii) would give $8\tau \sum_{\text{large}} \gamma < \tau \sum_N \gamma \le 8\tau \sum_{\text{large}} \gamma$; so the large γ vanish, and Lemma S4.4(i) gives $e_c \le 4\tau \cdot 4$. □

Lemma S4.4 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \le 4$. (ii) If $\sum_L \gamma_{ij} > 0$, then $\sum_N \gamma_{ij} \le 8 \sum_L \gamma_{ij}$. Both bounds are attained. *Proof.* In (i) $|\Delta k| \le 1$ on L, so all $|k_i| \le 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \le 15$; beyond, some $|k_i| \ge 16$, and since L connects the four vertices with $k_0 = 0$, $Q = \sum_L \Delta k^2 \ge 16^2/3 > 75$. For each pattern (L, N) let $\rho_{L,N}$ be the largest generalised eigenvalue of the quadratic form $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$; the largest over all pairs (L, N) is $3 + 2\sqrt{2}$. Then $\sum_N \gamma \le \rho_{L,N} Q + \sqrt{|N|\rho_{L,N} Q}$ and $\sum_L \gamma \ge Q - \sqrt{|L|Q}$, and the ratio of these bounds, $(\rho_{L,N}\sqrt{Q} + \sqrt{|N|\rho_{L,N}})/(\sqrt{Q} - \sqrt{|L|})$, decreases in Q, so it suffices to check it at $Q = 75$ for each pattern with its own $\rho_{L,N}$, where it is below 8 in every case; with the uniform value $3 + 2\sqrt{2}$ it would not be. The lemma is computer-assisted. Equality in (ii): $L = \{01, 02, 13\}$, $N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$. □

Theorem S4.5. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) = \{F \in \mathcal{C}(\Lambda^*) : p_{ij}(F; \mathbf{G}) \ge -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma S4.3 gives $e_c(F; \mathbf{G}) \le 16\tau$; the excess for Λ^* is the same sum with the zeroed slots dropped, so $e_c(F; \Lambda^*) \le 16\tau < 4m^*$, and Lemma S4.2 makes F obtuse for Λ^*. □

The operators from F_0 to the members of $\mathcal{C}(\Lambda^*)$ are integer matrices independent of

the metric, so one enumeration serves every band up to the one idealised at.

Lemma S4.6 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member of the closure for Z' is a member of the closure for Z. *Proof.* The members for Z' are reached from F_0 by words of steps, each at a slot that is zero at that point of the word (Lemma A3 of the main text); a step at a slot zero for both lattices acts on both conorm vectors as the same transposition of two slots, so after each step the superbase is obtuse for both and the next step is again at a slot zero for both. □

Proposition S4.7 (reindexing guarantee in the reference basis). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference's reduced basis, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \leq \xi < \nu_{\min}$ with $\nu_{\min}$ the smallest eigenvalue of $\mathbf{G}_R$, let F_N be its reduced superbase with basis vectors of summed squared length V, and let $\tau \geq \sqrt{2}\,\xi\,V/(\nu_{\min} - \xi)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, with its true integer coefficients in the reference basis, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* With u_1, u_2, u_3 the coordinate vectors of F_N's basis in the reference basis and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$, $\sum |u_i|^2 \leq V/(\nu_{\min} - \xi)$; in F_N's own basis the error has Frobenius norm at most $\sum |u_i|^2\,\xi$, and every conorm shift is at most $\sqrt{2}$ times that norm. The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem S4.5 applies. □ For fifteen references near zero conorms of five zero patterns, each with twelve new cells carrying a random metric error and a random reindexing, the true operator was among the candidates in 180 of 180 cases at the band given by the coverage bound below; the guarantee of Proposition S4.7 is conservative, a median 13ξ against a band near 3ξ that was enough.

From a metric error to a band. A metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by $-u_i^{\mathsf{T}}\Delta\mathbf{G}\,u_j$, at most $c_{ij}\xi$ with $c_{ij} = \|\operatorname{sym}(u_i u_j^{\mathsf{T}})\|_F$; every member survives every error of norm ξ exactly when $\tau \geq \tau_{\text{cov}} = \max(c_{ij}\xi - p_{ij})$, read off the stored closure. The largest constant on zero conorms is $\sqrt{2}$ with one zero,

$\sqrt{5}$ with two, and $2\sqrt{3}$ in six of the sixteen labelled cuboid bases, so $\tau = 3\xi$ is not a worst-case guarantee in every basis. These basis-dependent constants are what the strain norm removes.

S5. Further tables

Table S5. *The seven sample cells of BLDConograph (Oishi-Tomiyasu, 2024): the program's Bravais type and its minimal distance to that type (Å^2), and the same divided by the mean squared edge (%); the Bravais types of the sort spectrum in increasing strain distance (%), aP omitted, up to and including the program's type; and the next type with its distance. For all seven cells r_1 reaches the cap, so "> 25" is a lower bound on the distance as well as the outcome of the minimum over the bases of $\mathcal{U}$. A band of 7.4% selects the program's type in all seven; the largest ratio between consecutive distances follows it in six, and for the alumina cell lies between mC and hR.*

sample	type	Å^2	%	types up to the program's (strain %)	next
HERMES_Sr327_250K	tI	0.57	1.24	mC 0.31, oI 0.74, oF 0.83, tI 0.84	mP > 25
1339prbasrcuo_bs1	tP	0.056	0.10	mP 0.11, mC 0.14, oS 0.17, oP 0.18, tP 0.22	oF > 25
sample3	cI	2.07	0.78	mC 0.23, hR 0.39, oF 0.39, oI 0.49, tI 0.49, cI 0.58	cF > 25
CeO2_matsushita	cF	0.084	0.61	mC 0.09, oI 0.17, oF 0.17, tI 0.18, hR 0.26, cF 0.30	cI > 25
M402-pf-119673	hP	20.5	5.28	mC 0.99, mP 2.24, oS 2.37, hP 2.94	oI > 25
Acrinol-B-D8CuKa1	mC	0.72	0.35	mC 0.31	oF 20.6
Alumina676	hR	1.93	8.11	mC 0.43, hR 3.58	oI 14.9

Table S6. *The widened closure of the measured worked cell. Each row is a member, named by a word in the steps of Table 1 of the main text from the reduced cell e (the empty word); the six numbers are its conorms under the measured metric, in Å^2; "depth" is the strain at which it enters, in %. At a strain of* 7.7%*, the widening for the default band, all sixteen are members.*

word	*da*	*db*	*dc*	*ab*	*ac*	*bc*	depth
e	49.5	1588.9	2444.7	4.2	847.5	4.5	0
bc	45.0	8.7	2449.3	1593.4	852.0	−4.5	0.28
ab	851.7	1593.1	2440.6	−4.2	53.7	8.7	0.49
bc.db	2458.0	−8.7	53.7	1602.1	843.3	4.2	0.62
ab.bc	842.9	4.5	2449.3	1601.8	62.5	−8.7	0.78
ab.bc.db	2453.8	−4.5	847.5	1606.3	57.9	−4.2	0.78
da	−49.5	2494.3	1638.4	53.7	897.0	−45.0	3.76
ab.bc.ac	1664.2	−57.9	2511.7	905.4	−62.5	53.7	4.56
da.ab	950.7	2548.0	1584.7	−53.7	4.2	8.7	5.06
da.ab.ac	−49.5	2543.8	1588.9	954.9	−4.2	12.9	5.08
ab.ac	49.5	1539.3	2494.3	905.4	−53.7	62.5	5.08
da.ab.bc	942.0	−45.0	1593.4	2556.7	12.9	−8.7	5.09
bc.db.dc	45.0	2511.7	−53.7	1548.4	897.0	57.9	5.09
ab.ac.bc.ac	1610.5	959.1	2565.5	−4.2	−8.7	−53.7	5.11
da.ab.bc.ac	2569.6	−57.9	1606.3	954.9	−12.9	4.2	5.11
ab.ac.bc	−12.9	967.8	2556.7	1601.8	8.7	−62.5	5.11

Table S7. *The sort patterns of Andrews & Bernstein (2023), in the S^6 slot order bc, ac, ab, ad, bd, cd, and the same patterns in the slot order da, db, dc, ab, ac, bc of Table S10. The representative used here is their pattern read in the other order, with the sign of the Selling parameters reversed; no reflection is applied. Which zeros a pattern has also fixes the kind of face: for the three-zero sorts C5, T5 and O5 the zeros are ab, ac and bc, a triangle on a, b, c, and the same lattices also have reduced cells whose zeros form a path.*

sort	S^6 order	this paper	sort	S^6 order	this paper
C1	rrrrrr	rrrrrr	O2	rs0srt	srt0sr
C3	rr0rr0	rr00rr	O3	rs0rs0	rs00sr
C5	000rrr	rrr000	O4	00rsst	sstr00
H4	00rrrs	rrsr00	O5	000rst	rst000
T1	rrsrrs	rrssrr	M1A	rrsttu	ttusrr
T2	rr0rrs	rrs0rr	M1B	rstrsu	rsutsr
T5	000rrs	rrs000	M2A	rs0stu	stu0sr
R1	rrrsss	sssrrr	M2B	rs0rst	rst0sr
R3	rr0sr0	sr00rr	M3	rs0ts0	ts00sr
O1A	rrsrrt	rrtsrr	M4	00rstu	stur00
O1B	rstrst	rsttsr	A1	rstuvw	uvwtsr
			A2	rs0tuv	tuv0sr
			A3	rs0tu0	tu00sr

Table S8. *Pseudo-symmetric PDB entries (Le Page lattice above the reported type at 3°) by the Delone sort of the Le Page lattice, operators of the reported lattice's own holohedry excluded, entries with an exact higher lattice included (their smallest price is 0): number of entries, and the 10th/50th/90th percentiles of the largest obliquity δ (degrees) and of the largest and smallest price (%) among the pseudo-symmetry operators of each entry. Sorts with no entries are omitted. The transitions from the reported type are in Table S9. From* `results_pdb_lepage.txt`.

sort	type	entries	δ (°)	largest price (%)	smallest price (%)
C1	cI	593	0.14/1.64/2.73	0.69/8.10/13.58	0.32/7.13/13.18
C3	cF	562	0.19/1.72/2.76	0.95/8.51/13.61	0.50/7.68/13.41
C5	cP	1 349	0.29/1.70/2.72	1.43/8.40/13.45	0.47/5.52/12.59
H4	hP	2 008	0.08/1.40/2.70	0.40/6.91/13.32	0.11/4.03/12.70
T1	tI	1 376	0.06/1.32/2.67	0.28/6.60/13.33	0.17/5.55/12.60
T2	tI	330	0.05/1.48/2.69	0.24/7.30/13.43	0.11/4.98/12.44
T5	tP	14 816	0.22/1.47/2.69	1.07/7.27/13.43	0.70/6.56/12.98
R1	hR	96	0.04/1.55/2.76	0.22/7.64/13.78	0.13/2.98/13.13
R3	hR	393	0.07/1.82/2.81	0.37/9.09/13.86	0.29/7.64/13.72
O1A	oF	1 126	0.23/1.51/2.70	1.12/7.45/13.33	1.02/7.05/13.17
O1B	oI	331	0.10/1.58/2.78	0.49/7.80/13.73	0.36/7.63/13.73
O2	oI	1 078	0.14/1.40/2.73	0.70/6.92/13.50	0.48/6.33/13.36
O4	oS	11 679	0.14/1.49/2.74	0.67/7.36/13.56	0.62/7.15/13.51
O5	oP	4 760	0.03/1.08/2.59	0.17/5.33/12.79	0.15/5.13/12.69
M1A	mC	867	0.72/2.09/2.83	3.53/10.33/13.98	3.53/10.33/13.98
M1B	mC	398	0.12/1.80/2.72	0.57/8.91/13.41	0.57/8.91/13.41
M2A	mC	854	0.13/1.75/2.77	0.66/8.64/13.66	0.66/8.64/13.66
M2B	mC	2	0.61/0.95/1.29	3.02/4.70/6.38	3.02/4.70/6.38
M3	mC	1	2.39/2.39/2.39	11.80/11.80/11.80	11.80/11.80/11.80
M4	mP	582	0.10/1.40/2.65	0.47/6.89/13.07	0.47/6.89/13.07

Table S9. *The cheapest pseudo-symmetric alternative of PDB entries without an exact higher lattice (operators of the reported lattice's own holohedry excluded), by transition from the reported Bravais type to the one Le Page's* 3° *finds: entries with priced alternatives, the median price (%), and the share within* 1%*, within the 90th percentile of non-isomorphism (*4.2%*) and within its 95th (*6.7%*). The sixteen most frequent transitions.*

transition	entries	median	≤ 1%	≤ 4.2%	≤ 6.7%
oP→tP	12 120	6.98	9	31	48
mP→oS	9 474	7.38	12	33	46
mP→oP	4 417	5.78	20	42	55
aP→mC	2 113	9.49	9	19	30
mC→oS	1 831	7.41	12	33	45
mP→tP	1 542	3.66	27	54	69
mC→oI	1 308	7.13	11	30	48
oS→hP	1 169	7.34	10	26	45
oI→tI	1 165	6.25	19	38	52
mC→oF	1 067	7.29	8	29	46
oS→tP	880	7.42	14	32	46
tP→cP	724	7.10	7	28	47
aP→mP	571	6.97	15	30	48
mP→hP	479	0.71	61	91	100
mC→hR	391	8.92	19	29	37
mC→tI	391	4.56	23	46	64
all	42 656	6.69	14	35	50

Over the sorts with more than two entries, the median largest price lies between 5.3% (O5, oP) and 10.3% (M1A); the median largest obliquity between 1.1° and 2.1°. Within a sort the median largest price of the operators of order 3, 4 or 6 is within half a percentage point of that of the twofolds (`results_pdb_lepage.txt`).

The closures of the 24 sorts. For each face the matrices are the operators of $\mathcal{U}$ that map the face into the cone; on the face every matrix permutes the six conorm slots, so each fixes a subspace of conorm vectors, and the sorts are the closed sets of equalities obtained by intersecting those subspaces (Definition A5 of the main text). For a sort, the point group is the set of matrices that fix the sort's subspace pointwise, the visible symmetries are those that are relabellings of the reduced cell, and two superbases have the same shape when their conorms agree up to a slot permutation induced by a relabelling. Table S10 is the resulting catalogue; every entry is the result of this linear algebra on integer matrices. In every sort the number of symmetries out of each labelled superbase is the order of the holohedry, and the twofolds, with the inversion,

generate the group.

Symmetry visible in one cell. In the sorts with zero conorms the symmetries that relabel the reduced cell are a minority: 2 of 8 for primitive orthorhombic, 4 of 16 for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal. None of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell, nor the one twofold of primitive monoclinic. The orbit of the reduced cell under the point group has |group|/visible superbases, all of its shape: for oP the orthogonal cell and three monoclinic-looking cells with two right angles, one for each axis traded for a face diagonal. Reduction of a measured orthorhombic cell stops at whichever of the sixteen superbases its errors favour; an orthogonal superbase is always one step away, but the one in the reference's labelling can be four steps away, and no twofold of the lattice reaches it. The catalogues of shapes and of the conditions under which one sort becomes another are Tables S1 and S2.

Table S10. *The Selling closure of each Delone sort, computed from the operators alone. Patterns are conorms in the slot order da, db, dc, ab, ac, bc; equal letters denote equal values and* 0 *a zero; they are the patterns of Andrews & Bernstein (2023) read in this order (Table S7). Superbases are counted up to relabellings. "Visible" is the number of symmetries that relabel the reduced superbase written in the pattern shown; another superbase of the same lattice can show a different number. "Shapes" is the number of distinct cell shapes among the superbases, equal in every case to the number of orbits of superbases under the symmetries.*

sort	Bravais	pattern	zeros	superbases	|group|	visible	shapes
C1	cI	rrrrrr	0	1	48	48	1
C3	cF	rr00rr	2	3	48	16	1
C5	cP	rrr000	3	16	48	12	2
H4	hP	rrsr00	2	6	24	4	1
T1	tI	rrssrr	0	1	16	16	1
T2	tI	rrs0rr	1	2	16	8	1
T5	tP	rrs000	3	16	16	4	3
R1	hR	sssrrr	0	1	12	12	1
R3	hR	sr00rr	2	3	12	4	1
O1A	oF	rrtsrr	0	1	8	8	1
O1B	oI	rsttsr	0	1	8	8	1
O2	oI	srt0sr	1	2	8	4	1
O3	oI	rs00sr	2	3	8	8	2
O4	oS	sstr00	2	6	8	4	2
O5	oP	rst000	3	16	8	2	4
M1A	mC	ttusrr	0	1	4	4	1
M1B	mC	rsutsr	0	1	4	4	1
M2A	mC	stu0sr	1	2	4	2	1
M2B	mC	rst0sr	1	2	4	4	2
M3	mC	ts00sr	2	3	4	4	2
M4	mP	stur00	2	6	4	2	3
A1	aP	uvwtsr	0	1	2	2	1
A2	aP	tuv0sr	1	2	2	2	2
A3	aP	tu00sr	2	3	2	2	3

S6. Distances to the 24 sorts

The same computation gives the distance from a measured cell to each of the 24 sorts. In each of the 6096 bases of $\mathcal{U}$ the cell's conorms are projected, in the strain, onto the sort's subspace (its zeros and equalities, Definition A5 of the main text, the other conorms non-negative), and the smallest distance is taken. This is the computation of SELLA (Andrews *et al.*, 2023), which projects the boundary images of a cell in S^6 onto the same subspaces, and of Oishi-Tomiyasu (2012) in Selling coordinates; projection onto Bravais-lattice subspaces goes back to Andrews & Bernstein (1988). What

changes is the norm, which is the same in every basis, and the set of bases, which is $\mathcal{U}$. We call the 24 distances the *sort spectrum* of the cell. A distance below the adjacency radius r_1 is exact, since the bases of $\mathcal{U}$ then contain the basis of the nearest lattice of the sort; a minimum above r_1 is an upper bound over the bases used, the true distance being at least r_1, and the implementation flags it (`certified_sort_spectrum`). Along every specialisation of Table S2 the distance cannot decrease, since the lower sort's subspace lies inside the upper one's.

Fig. S1 draws the spectrum of the worked cell on the graph of specialisations. The nearest sorts are A2 at 0.28%, M4 (mP) at 0.78%, A3 at 3.8% and O5 (oP) at 5.1%, and every other sort is at least 25% away, a bound that holds because r_1 reaches the cap for this cell. For a tetragonal lysozyme cell indexed in $P1$ the twelve sorts from aP to tP all lie within 0.22% (Fig. S2).

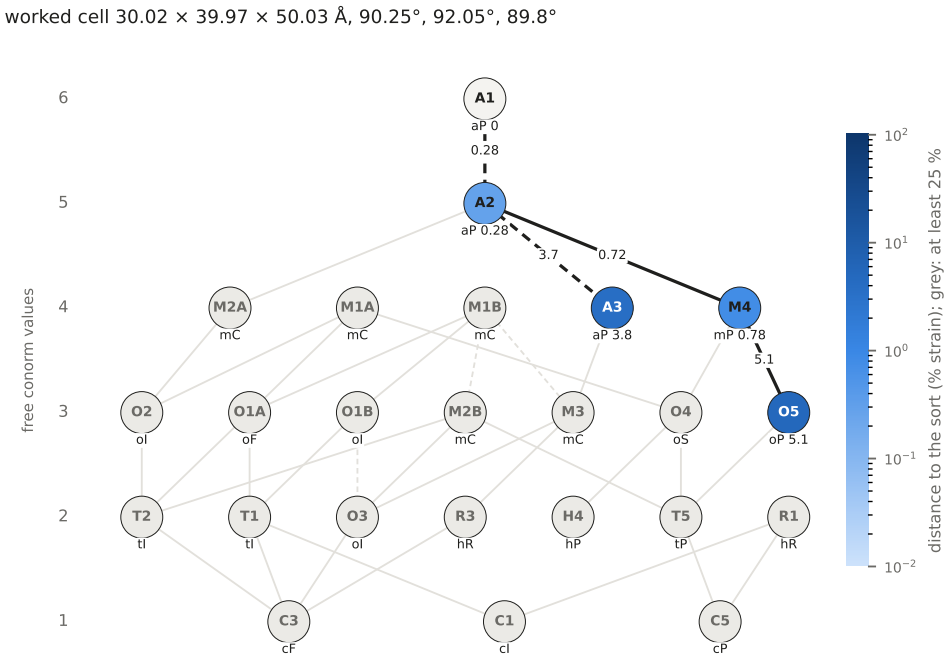


Fig. S1. The 24 Delone sorts as a graph for the measured worked cell. Rows are the number of free conorm values. Each node gives the Bravais type and the strain distance (%) from the cell to the sort, its colour the same distance on a log scale; sorts at least 25% away are grey. Edges are the specialisations of Table S2, labelled with the distance from the cell idealised onto the upper sort to the lower one. Dark edges are routes on which these steps add in quadrature to the lower sort's distance within 3%, drawn where that distance is below 10%. Dashed edges keep the Bravais type.

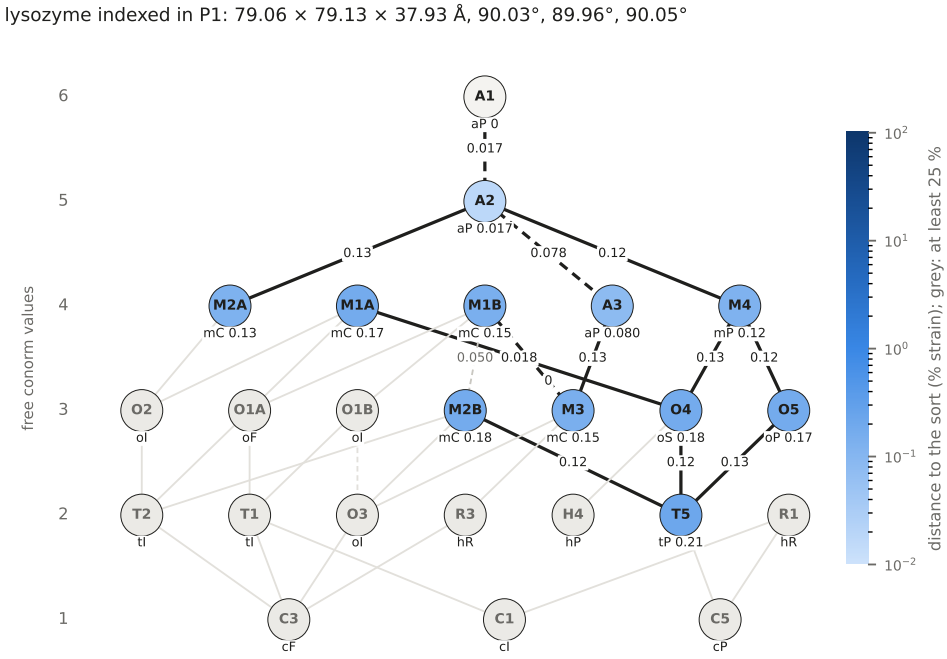


Fig. S2. The 24 Delone sorts as a graph for a tetragonal lysozyme cell (79.1 × 79.1 × 37.9 Å) indexed in $P1$ with an added error, drawn as Fig. S1. The twelve sorts from aP to tP lie within 0.22%; every other sort is at least 25% away (grey).

Le Page's 3° *by sort.* The distance to a sort at Le Page's boundary depends on the sort, while the price of each operator does not. For exact lattices of each sort perturbed in random directions until Le Page's obliquity for the sort's Bravais group is exactly 3°, the median largest twofold price is 14.8% in every sort, which Proposition A9 makes true by construction and the table confirms numerically (larger only in the monoclinic tails, up to 40–55%, where Le Page's group is another C-centred group), while the median distance to the sort is 7.4–7.7% for the orthorhombic and monoclinic sorts other than O3, M2B and M3, 7.9–8.1% for tP, tI, hP and hR, and 8.3–8.5% for the cubic sorts, with longer tails for the boundary sorts O3, M2B and M3 (Table S11). A single band on the distance to a sort therefore tests each sort at a different level, which is why the reindexing in this paper is decided on the prices of operators.

Table S11. *Le Page's* 3° *by Delone sort: for exact lattices of each sort, each perturbed in a random direction until the obliquity of the sort's Bravais group is* 3°*, the strain distance to the sort (median, and 10th–90th percentiles) and the median of the largest price of the lattice's twofolds and of its other rotations, by order, in %. In the monoclinic sorts some perturbations reach the boundary through a different C-centred group, whose own twofold costs* 14.8% *while the exact lattice's costs more. Of 25 directions per sort, those that never reach an obliquity of* 3° *are dropped, hence* $n < 25$ *in seven sorts; with* $n \leq 25$ *the percentiles are indicative.*

sort	type	n	distance to sort	twofold price	other orders
C1	cI	25	8.3 (7.6–9.0)	14.8	3: 14.1, 4: 15.0
C3	cF	25	8.3 (7.7–9.2)	14.8	3: 14.3, 4: 14.5
C5	cP	25	8.5 (7.6–9.3)	14.8	3: 14.2, 4: 14.7
H4	hP	25	7.9 (7.5–8.5)	14.8	3: 13.7, 6: 11.3
T1	tI	25	7.9 (7.5–9.1)	14.8	4: 14.0
T2	tI	25	8.0 (7.6–8.8)	14.8	4: 14.4
T5	tP	25	8.0 (7.4–8.9)	14.8	4: 14.1
R1	hR	25	8.0 (7.5–8.6)	14.8	3: 14.4
R3	hR	25	8.1 (7.5–8.8)	14.8	3: 14.5
O1A	oF	24	7.5 (7.4–7.8)	14.8	—
O1B	oI	25	7.6 (7.4–8.1)	14.8	—
O2	oI	25	7.6 (7.4–8.2)	14.8	—
O3	oI	25	8.6 (7.6–13.6)	14.8	—
O4	oS	25	7.7 (7.4–8.2)	14.8	—
O5	oP	25	7.5 (7.4–8.3)	14.8	—
M1A	mC	20	7.4 (7.4–9.1)	14.8	—
M1B	mC	15	7.4 (7.4–7.4)	14.8	—
M2A	mC	18	7.4 (7.4–8.5)	14.8	—
M2B	mC	17	9.3 (7.5–19.9)	14.8	—
M3	mC	23	8.9 (7.5–16.1)	14.8	—
M4	mP	23	7.4 (7.4–7.4)	14.8	—

Against BLDConograph and SELLA. On the seven sample cells that Oishi-Tomiyasu (2024) distributes with BLDConograph, the highest Bravais type within 7.4% is the program's type in all seven, with the next type at least 14.9% away (Table S5). SELLA (`CmdSella`, Andrews *et al.*, 2023, default settings) was run on the worked cell, the lysozyme cell, a pseudo-tetragonal cell, a cell with all angles 92°, a needle, two plates and the seven BLDConograph cells. The two rankings of the 24 sorts agree closely, with Spearman's rank correlation 0.77 to 0.97 over all 24 sorts, which the many distant sorts inflate. The two scales are not proportional: SELLA's distance divided by the cell's mean squared edge is a median 0.71 times the strain distance, with a 10th to 90th percentile range of 0.05 to 26 (`results_sella.txt`). With SELLA's band set

by that median ratio, which is calibrated on the same 14 cells and so flatters the agreement, the Bravais types within the band agree in 11 of the 14 cells. They differ on the needle and the two plates, where the normalisation by the mean squared edge, our choice and not SELLA's, weighs the long edges differently: SELLA admits hP, hR, oF, tI and tP for the needle, which are far in strain, and the strain band admits oF (and tI) for the plates, which SELLA's does not. For these thin cells r_1 is 4 to 6%, so their strain distances above it are bounds and are flagged.

S7. Computational checks

Every check is a function of the package `sellingproofs` in the directory `manuscript/supplement` of the `agentsg` repository. A decorator, `@claim`, registers each function with the number *znn* of the statement it verifies, the places in the paper where the statement is made, and its kind: a machine-checked proof (formal), an exhaustive finite computation (exhaustive), spot checks on seeded samples (sampled), an exact rational certificate (certified), a number recomputed from the deposited results (benchmark, pdb), or a rerun of a script that regenerates its deposited results file (reproduce). The first paragraph of each function's documentation states the claim as the paper makes it. `python -m sellingproofs ledger` lists the claims and `python -m sellingproofs run` runs them as pytest tests; Table S13 is generated from the ledger. The statement groups are: z36: metric errors as conorm bands, coverage constants (4); z44: the band theorem: class excess, generation by flips, brute force (5); z45: the worked cell's widened set, walks and reduction under a band (4); z46: the closure groupoid of every sort, the monoclinic ring (7); z47: reindexing under a declared metric error (1); z48: Delone's sorts from the groupoid (1); z49: bands per slot, cell shapes as classical ambiguities (2); z50: codimension-one specialisations (1); z51: the BLDConograph samples (1); z52: the operator set U: 6096 operators, kinds, entries (1); z53: sorts as closed sub-

spaces (1); z54: the table of sorts from the operators (1); z55: exactness beyond the proved band (1); z56: the window lemma on all 38 faces, completeness of U (3); z57: bands per slot on needles and plates (5); z58: projection, enumeration and filtering, needles and plates (4); z59: balls, layers and adjacency radii (2); z60: strains, prices, symmetries and the worked cell (16); z61: the appendix statements: Voronoi characterisation, generation, A4, A10, A11 (8); z62: the machine-checked proofs (Lean) (31); z63: the implementation (library tests) (18); z64: the benchmark numbers and their reproduction (44); z65: the PDB numbers and their reproduction (25); z66: remaining statements, tables and figures (26); z67: the reindexing workflow: identities, worked example, comparison with the direct form (6); z68: the Niggli window and the way from a reduced cell to the Niggli cell (8). `python -m sellingproofs pin` prints the commit, the `agentsg` version (0.5.0) and the SHA-256 hashes of the two copies of $\mathcal{U}$, which a claim checks. The checks were run with Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1 and cctbx-base 2025.11. A claim whose data or programs are absent is skipped with the reason, or fails when asked to (`--strict-needs`): the PDB claims need the Zenodo snapshot (doi:10.5281/zenodo.22986222) in `data/`, the comparisons need cctbx and SELLA (CmdSella), and the formal claims Lean.

Formal proofs. The Lean development in `manuscript/supplement/formal` builds with Lean 4.35.0 and Mathlib release v4.35.0 (`lake build`). Its data file is generated from `universal_operators_columns.json` and records the file's SHA-256 hash, which `check_data.py` recomputes. `audit/axioms.lean` prints the axioms of each headline theorem. Table S12 lists the theorems with the statements they prove. Beyond the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`), the completeness of $\mathcal{U}$ (`arrow_mem_U`, and `arrow_mem_U_of_posdef` for every lattice) depends on one compiled evaluation (`native_decide`, of `finite_check`); its converse (`U_arrow`) on two (`U_realised`, `U_unimodular`); the dependence on the zero pattern

(`arrow_zero_pattern` and `arrow_zero_pattern_of_posdef`) on two (`finite_check`, `zero_pattern_check`); the size of $\mathcal{U}$, its distinct elements, its entries, the 38 zero patterns, and the closure of $\mathcal{U}$ under inversion and under the 48 relabellings on one each. That the non-zero conorms of a lattice connect the four vectors (`conn_of_posdef`), the window (`box_of_conn`, Lemma A2 of the main text), Lemma A1 (`excess_identity`), the shortest partial sums (`partial_shortest`), every part of Propositions A10 and A9(i) (`price_mul`, `price_inv`, `coset_lower`, `coset_upper`, `Hgroup`, `candidates_eq_coset`, `stab_sub_H`, `H_sub_H`, `H_eq_stab`, `price_le_two_dist_fixed`, `twofold_midpoint_fixed`) and Proposition A11(i)–(ii) depend on the standard axioms alone, that is, on the kernel.

What the formal statements assume. The formal statements are narrower than, or equal to, the prose in the following ways, each of which the prose states or the text bridges. (i) Lattice vectors are coordinates in the basis (a, b, c) of a reduced superbase with conorms $p \geq 0$, and dot products are given by Selling's identity, so every statement about $\mathcal{U}$ is in the basis of the first superbase, which is how $\mathcal{U}$ is defined; since any obtuse superbase can serve as the first, this covers the matrices between any two. (ii) A lattice is a metric positive definite on the non-zero integer vectors; `conn_of_posdef` derives from it the connected support that `arrow_mem_U` assumes. (iii) The window is $[-7, 7]^3$, wider than the $[-6, 7]$ of the prose. (iv) Of the characterisation of obtuse superbases by shortest partial sums, the forward direction is proved for every metric (`partial_shortest`); the converse, that a superbase whose partial sums are all shortest on a pattern is obtuse there, is checked for the elements of $\mathcal{U}$ by `zero_pattern_check`, which is all the completeness proof needs. (v) Propositions A9(i) and A10 are proved for any isometric action of a group on a metric space, with the price the displacement $D(x, g \cdot x)$; they apply to the paper's prices because a unimodular change of basis is an isometry of the affine-invariant distance between metrics normalised to

unit determinant (Section 2.4 of the main text), which is not formalised. The order of products follows the left action $g \cdot \mathbf{G} = g^{-\mathsf{T}}\mathbf{G}g^{-1}$, under which the paper's $h\mathbf{P}_0$ is written $\mathbf{P}_0 h$. (vi) The equality case of Proposition A9(i) assumes that the midpoint of $\mathbf{G}$ and $g^{\mathsf{T}}\mathbf{G}g$ exists and is unique, which holds in the space of positive definite matrices (Bhatia, 2007) but is not formalised. (vii) Proposition A11 is formalised in the Gram entries of the cell, with Niggli's main conditions only, which is how it is stated.

Table S12. *The machine-checked theorems (Lean 4 with Mathlib, in* `manuscript/supplement/formal`*): the statement each proves, where the paper makes it, and the compiled evaluations (*`native_decide`*) it rests on beyond the three standard axioms. Generated from the package's ledger (claim z62), which builds the development and checks each theorem's axioms.*

theorem	place	statement	compiled evaluations
`arrow_mem_U`	§3.3; Lemma A2	Completeness of $\mathcal{U}$ for conorms $p \geq 0$ with connected support: every unimodular matrix from that superbase to another obtuse superbase of the metric, in the basis of the first, is in the deposited table.	`finite_check`
`arrow_mem_U_of_posdef`	§3.3; Lemma A2; Table 6	Completeness of $\mathcal{U}$ for every lattice (positive definite metric).	`finite_check`
`conn_of_posdef`	Lemma A2; SI §S12	The non-zero conorms of a positive definite metric connect the four vectors of its reduced superbase.	none
`arrow_zero_pattern_of_posdef`	§3.3; Lemma A1	For every lattice, arrows depend only on the zero pattern.	`finite_check`, `zero_pattern_check`
`U_length`	§3.3; Appendix A, the enumeration	$\lvert\mathcal{U}\rvert = 6096$.	`U_length`
`U_nodup`	§3.3	The elements of $\mathcal{U}$ are distinct.	`U_nodup`
`U_entries`	§3.3; Table 4	Every entry of $\mathcal{U}$ is in $\{-3, \ldots, 3\}$, and some is ± 3.	`U_entries`
`faces_count`	§2.1; Appendix A, the enumeration	There are 38 zero patterns.	`faces_count`
`U_arrow`	§3.3	Every element of $\mathcal{U}$ is an arrow of every lattice on some face.	`U_realised`, `U_unimodular`
`arrow_zero_pattern`	§3.3; §3.2	Arrows depend only on the zero pattern: for conorms with connected support, an arrow for one metric is an arrow for every metric with the same zero conorms.	`finite_check`, `zero_pattern_check`
`box_of_conn`	Lemma A2	The window of Lemma A2: $\gamma = 0$ on a connected support puts a vector in $[-7, 7]^3$.	none
`U_swaps`	Remark A8	$\mathcal{U}$ is closed under the 48 swaps.	`U_swaps`
`U_inv`	§3.3	$\mathcal{U}$ is closed under inversion.	`U_inv`
`excess_identity`	Lemma A1	The class excess identity of Lemma A1.	none
`partial_shortest`	SI §S12, Terms	The partial sums of an obtuse superbase are shortest in their parity classes.	none
`price_mul`	Proposition A10(i)	$\pi(h_1 h_2) \leq \pi(h_1) + \pi(h_2)$.	none
`price_inv`	Proposition A10(i)	$\pi(h^{-1}) = \pi(h)$.	none
`coset_lower`	Proposition A10(ii)	$H_{\epsilon - p_0}\mathbf{P}_0 \subseteq C_\epsilon$.	none
`coset_upper`	Proposition A10(ii)	$C_\epsilon \subseteq H_{\epsilon + p_0}\mathbf{P}_0$.	none
`Hgroup`	Proposition A10(iii)	With no price in $(\epsilon, 2\epsilon]$, H_ϵ is a group.	none
`candidates_eq_coset`	Proposition A10(iv)	With no price in $(\epsilon - p_0, \epsilon + p_0]$, $C_\epsilon = H_\epsilon\mathbf{P}_0$.	none
`stab_sub_H`	Proposition A10(v)	$\mathrm{Aut}(\mathbf{G}_0) \subseteq H_{2\nu}(\mathbf{G}_R)$.	none
`H_sub_H`	Proposition A10(v)	$H_\epsilon(\mathbf{G}_R) \subseteq H_{\epsilon+2\nu}(\mathbf{G}_0)$.	none
`H_eq_stab`	Proposition A10(v)	With a gap above $\epsilon + 2\nu$ and $\epsilon \geq 2\nu$, $H_\epsilon(\mathbf{G}_R) = \mathrm{Aut}(\mathbf{G}_0)$.	none
`twofold_midpoint_fixed`	Proposition A9(i)	For a twofold g, the midpoint of x and $g \cdot x$ is fixed by g.	none
`price_le_two_dist_fixed`	Proposition A9(i)	The price of g is at most twice the distance to the metrics g fixes.	none
`Niggli.typeII_obtuse`	Proposition A11(i)	A type II Niggli cell with $d = -a - b - c$ is an obtuse superbase.	none
`Niggli.typeII_d_conorms`	Proposition A11(i)	For a type II Niggli cell, $p_{dx} \geq \frac{1}{2}x \cdot x - \min(\lvert x \cdot y\rvert, \lvert x \cdot z\rvert)$.	none
`Niggli.typeII_separation`	Proposition A11(i)	Every conorm on d exceeds every conorm off d that is at most $\tau < \frac{1}{4}a \cdot a$.	none
`Niggli.typeI_no_orthogonal_obtuse`	Proposition A11(ii)	For a type I cell none of the four orthogonal superbases is obtuse.	none

Certified radii. The depth of a superbase F with conorms q under the metric is the least strain norm $\|r\|_M$ of a change $r \geq -q$ of its conorms, with M the strain norm in the basis of F ($r^{\mathsf{T}} M r = \mathrm{tr}(\mathbf{G}^{-1}\Delta\mathbf{G}\,\mathbf{G}^{-1}\Delta\mathbf{G})$). For any multipliers $\lambda \geq 0$ with $c = -\lambda \cdot q > 0$ the depth is at least $c/(\lambda^{\mathsf{T}} M^{-1} \lambda)^{1/2}$, by weak duality, and any r with $q + r \geq 0$ whose metric is positive definite bounds it from above. `certify_radii.py` evaluates the balls of Lemma A6 and the conorm test of Proposition A4 exactly, so that every superbase within the cap is among its candidates, proves a lower bound for every candidate outside the layer and an upper bound for the nearest, all in rational arithmetic; floating point only proposes λ and r. The certificates are for the reduced conorms as stored in double precision, read as exact rationals, which agree with the cells of Table 8 to about 15 digits.

Timings. On a tenth of the benchmark, widening a reference took a median of 0.02 s and reindexing one new cell 0.03 s (90th percentiles 0.03 and 0.12 s); a $150 \times 120 \times 4.8$ Å plate took 0.9 s to widen. A $300 \times 300 \times 3$ Å plate takes about a minute to widen and its r_1 several minutes, and thinner cells are refused by the cap on the ball. A vectorised lookup of $\mathcal{U}$ for one new cell, reductions included, takes about 0.04 s (`adjacency_radii.py`).

Table S13. *The checks behind each place in the paper: the numbers znn of the statement groups whose checks cite it, and the number of checks. Generated by* `python -m sellingproofs tables` *from the ledger of the package; the subjects of the groups are listed in the text.*

main text	statements	checks
abstract	z64, z65	4
§2.1	z52, z56, z62	3
§2.2	z66	2
§3.1	z46, z54, z56	3
§3.2	z46, z62, z66	5
§3.3	z52, z56, z59, z61, z62, z63, z64, z66	18
§3.4	z60, z63	2
§4	z66	1
§4.1	z63, z66	2
§4.2	z60, z63, z66, z67	13
§4.3	z59, z60, z63, z65, z66	7
§4.4	z64, z65	5
§5.1	z45, z57, z58, z60, z63, z64, z66	26
§5.2	z60, z63, z64	6
§5.3	z65, z66	19
§5.4	z60, z64, z68	10
§6	z64, z65, z66	6
Table 1	z66	1
Table 2	z46	1
Table 4	z52, z56, z62	3
Table 5	z60, z63, z66	3
Table 6	z62, z63, z66	4
Table 7	z66	1
Table 8	z59, z63, z64	4
Table 9	z60, z63, z66	4
Table 10	z65	2
Fig. 1	z46, z66	2
Fig. 2	z60, z66	2
Fig. 3	z65, z66	3
Appendix A	z61, z62	3
Lemma A1	z44, z62	3
Lemma A2	z52, z56, z62, z64, z66	11
Lemma A3	z61	1
Proposition A4	z61	1
Definition A5	z53	1
Lemma A6	z59	1
Lemma A7	z60, z63	2
Remark A8	z59, z62, z63, z64, z66	8
Proposition A9	z60, z62, z64	7
Proposition A10	z60, z61, z62, z63, z66	14
Proposition A11	z60, z61, z62	6
Proposition A12	z68	5
Data availability	z66	1
Corollary A13	z63, z68	4

supporting information	statements	checks
§S2	z64	8
§S3	z52, z66	2
§S4	z36, z44, z45, z46, z49, z55, z57, z64, z66	23
§S5	z46, z48, z53	3
§S6	z58, z60, z63, z64	10
§S7	z62, z64, z66	6
§S8	z64, z65	3
§S9	z64, z65	9
§S10	z63, z67	7
§S11	z63, z68	5
§S12	z61, z62	5
Table S1	z46, z49, z66	3
Table S2	z50, z58	2
Table S3	z45	1
Table S4	z52	1
Table S5	z51, z60, z66	4
Table S6	z45, z60, z66	4
Table S7	z54	1
Table S8	z65	3
Table S9	z65	2
Table S10	z46, z54	2
Table S11	z64	2
Table S14	z64	5
Table S15	z65	3
Table S16	z67	1
Table S17	z67	1
Table S18	z67	1
Table S19	z67	1
Table S21	z68	1
Fig. S1	z60, z66	3
Fig. S2	z66	2
Lemmas S4.2	z44	1
Lemma S4.4	z44	1
Theorem S4.5	z44	2
Proposition S4.7	z47	1

S8. Existing searches on the same cases

The test cases of the main text were also given to the searches that data-processing programs use, each priced in the same way and kept within the same band: every matrix

with entries in $\{-1, 0, 1\}$ or $\{-2, \ldots, 2\}$ between the two Selling-reduced cells; the $\{-1, 0, 1\}$ search between the two Niggli cells; cctbx `similarity_transformations` between the two Niggli cells (Grosse-Kunstleve *et al.*, 2002) with generator ranges 1 and 2; and a pipeline that runs the cctbx lattice-symmetry analysis (Sauter *et al.*, 2004) on both cells and returns every operation of the reference's highest lattice group between the two standard settings. Table S14 gives the counts. The searches between Niggli cells and the cctbx similarity searches found the true operator in every case; the fixed windows between Selling-reduced cells, which no program uses, miss operators where Table 4 of the main text says they can; the pipeline missed a needle whose reference and measured cell idealise at 3° to different lattices, and a few stress and window cases. The purpose of the table is to show that the guarantee and the existing searches agree where both apply, not to rank them.

Table S14. *Existing searches on the test cases of the main text (Section 4.4): the true operator among the candidates, by method and family; candidates priced by the geodesic distance after the best scale and kept within a band of* 7.4% *(the reference widened at* 7.7% *for the closure and the lookup). Window family: cells measured in a member that needs an entry of* 3*, and near-mP and near-orthogonal lattices measured twice with an error of* 0.1 *to* 2%*, the reference given as its Niggli cell (split by Niggli type) or in a random basis (split by the kind of face). A dash: not run. The lookup on the benchmark and stress families is in* `results_lookup.txt`*. Results in* `manuscript/supplement/benchmark/`*.*

		stress			2- or 3-zero faces, reference given as Niggli cell		random basis	
method	benchmark	0.5ϵ	0.9ϵ	entry 3	type I	type II	2 zeros	3 zeros
closure	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
lookup of $\mathcal{U}$	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
$\{-1, 0, 1\}$, Selling cells	778/780	58/60	52/60	24/60	142/171	129/129	120/150	99/150
$\{-2, \ldots, 2\}$, Selling cells	780/780	59/60	60/60	52/60	170/171	129/129	150/150	149/150
$\{-1, 0, 1\}$, Niggli cells	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
cctbx similarity, range 1	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
cctbx similarity, range 2	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
lattice-symmetry pipeline	779/780	56/60	54/60	60/60	171/171	126/129	147/150	148/150

S9. Uncertainty of the PDB statistics, chance, and the Niggli window

Intervals. Table S15 gives block-bootstrap 95% intervals for the percentiles of Table 10 of the main text: the deposition blocks (runs of id-adjacent pairs) are resampled with replacement, 1000 times, which keeps the correlation within a deposition series. The shares of entries with an alternative within the band have intervals of ±0.2 to ±0.5 points, where a type with fewer than 200 blocks uses the pooled band (`pdb_band_uncertainty.py`).

Table S15. *Non-isomorphism in the PDB by reported Bravais type with block-bootstrap 95% intervals (cut* 20%*, identity as the operator, price after the scale fit, %).* [a] *Fewer than 200 deposition blocks; the pooled band stands in for these types in the shares of the main text, and the cubic rows are zero by construction.*

type	pairs	blocks	median	90th	95th
aP	902	535	2.06 (1.82–2.27)	7.66 (6.40–9.46)	11.68 (9.76–13.70)
mP	7 077	3 277	1.27 (1.21–1.33)	5.38 (5.01–5.84)	9.02 (8.21–9.78)
mC	4 932	1 977	1.49 (1.41–1.59)	5.06 (4.73–5.49)	8.40 (7.16–9.23)
oP	13 615	5 987	1.11 (1.07–1.14)	4.70 (4.46–4.95)	7.43 (7.06–7.81)
oS	2 271	1 108	1.01 (0.94–1.08)	3.57 (3.32–4.01)	5.50 (4.96–6.14)
oI	1 353	574	1.22 (1.07–1.39)	5.12 (4.28–6.16)	7.91 (6.40–9.15)
oF[a]	31	22	1.99 (1.23–3.10)	5.87 (3.39–8.28)	7.54 (4.48–8.95)
tP	5 174	2 061	0.52 (0.48–0.57)	2.59 (2.36–2.85)	4.00 (3.61–4.41)
tI	805	416	0.65 (0.53–0.78)	2.76 (2.28–3.34)	4.91 (3.61–6.03)
hR	1 270	597	0.68 (0.61–0.79)	3.54 (3.10–3.94)	5.86 (4.59–6.71)
hP	6 817	3 167	0.57 (0.53–0.62)	2.87 (2.65–3.04)	4.56 (4.22–4.95)
cP[a]	363	178	0.00 (0.00–0.00)	0.00 (0.00–0.00)	0.00 (0.00–0.00)
cI[a]	385	194	0.00 (0.00–0.00)	0.00 (0.00–0.00)	0.00 (0.00–0.00)
cF[a]	211	106	0.00 (0.00–0.00)	0.00 (0.00–0.00)	0.00 (0.00–0.00)
all	45 222	20 206	0.98 (0.96–1.00)	4.21 (4.09–4.33)	6.72 (6.52–6.98)

Against chance. Shuffling the free cell parameters independently among the entries of one space group keeps their distributions and destroys any relation between them; shuffling within twenty classes of cell volume keeps cells of one size together. At Le Page's 3° the real rate exceeds both shuffled ones for $P1$ (42–44% against 30–40%) and $P2_12_12_1$ (23–24% against 13–17%), whose real cells are more often pseudo-tetragonal, exceeds them slightly for $P2_1$ (46% against 39–44%), and matches them for $C2$ and $P2_12_12$. Near-exact pseudo-symmetry is not chance: the share whose cheapest alternative costs at most 1% is 10% for real $P1$ cells against 2–3% shuffled, and 9% against 5–6% for $P2_1$ (intervals about ±1 point). At 4.2% the excess remains for $P1$ (15–16% against 6–9%) but not clearly for $P2_1$ (18% against 13–18%, intervals about ±2

points), and mP entries carry much of the share within 4.2% (two draws of 1500 cells per arm; `results_pdb_lepage_null.txt`).

The crystal-form proxy. Random pairs of entries with one space group whose identifiers do not share their first three characters, drawn with the space-group mix of the id-adjacent pairs, pass the 20% cut in 1.5% of 19 986 cases (0.75% at 5%; `pdb_proxy_check.py`). Unrelated crystals rarely pass the cut by chance, so the movement of the percentiles with the cut reflects variation within deposition series rather than unrelated pairs.

The Niggli window. `niggli_window.py` draws 100 lattices on each of the 38 faces (half of them anisotropic, with vertex scales up to 30) and on each of the 24 sorts, measures each twice with independent errors of total strain 0.5, 2, 5 and 7.7%, the new cell in a random basis, and computes the true operator between the two Niggli cells (cctbx) and between the two Selling-reduced cells directly. Between Niggli cells its entries were in $\{-1, 0, 1\}$ in all 24 800 cases, strictly and not only up to the lattice's point group, and in 3 040 further cases on strongly anisotropic lattices (`niggli_window.py thin`: 40 per face, vertex scales up to 60, the square root of the metric's eigenvalue ratio up to 98, median 5); between Selling-reduced cells an entry of 2 or 3 was needed in 3 765 cases (on two zeros sharing a vector, three zeros, and the sorts cP, tP and oP) and in none once composed with the point group. Proposition A12 of the main text turns the first observation into a theorem for exact lattices and for measurements within a radius ρ of one lattice (Section S11).

S10. The reindexing workflow in detail

This section sets out, step by step, how the candidates of Section 4.2 of the main text are computed for a new cell once a reference has been widened, why each step is exact, what it costs, and what it gives on the worked cell. The procedure prices

exactly the operators that the direct form of Table 7 of the main text prices, the operator $\mathbf{P} = B_R \mathbf{W}_k \mathbf{Q} B_N^{-1}$ for every member $\mathbf{W}_k$ of the reference's widened set and every relabelling $\mathbf{Q}$; it differs only in where the work is done. Four observations reorganise it. (i) The two input bases drop out of every price, so each candidate can be priced in the new cell's reduced basis from a metric that depends only on the reference. (ii) Of the 48 relabellings of a superbase, two that differ by a common sign give the same metric, and the determinant rule keeps exactly one of them, so there are 24 metrics per member. (iii) Every candidate has the same best scale. (iv) A necessary condition on ten dot products of the new cell's reduced superbase, which follows from Lemma A7(i) and Proposition A4 of the main text, discards almost every ordering before an eigenvalue is computed. The exact operator is then built only for the candidates that lie within the band. The workflow is implemented in `agentsg` as the reindexing routine; a second entry point, `reindex_workflow`, returns every intermediate quantity named below, and the numbers of this section are produced by `reindex_workflow.py` (`results_reindex_workflow.txt`) and checked by claim z67 and by the library test registered as z63 (Section S7).

S10.1. Notation

The reference R is given as a cell and a space group. Its primitive cell is Selling-reduced (step 1 of Table 7 of the main text), which gives the reduced basis B_R, a 3×3 matrix whose columns are the reduced basis vectors a, b, c written in the reference's input basis, and the reduced metric

$$\mathbf{G}_R = B_R^{\mathsf{T}} \, \mathbf{G}_R^{\mathrm{in}} \, B_R, \tag{2}$$

with $\mathbf{G}_R^{\mathrm{in}}$ the metric of the input cell. B_R is integer for a primitive input cell and rational, with determinant $\pm 1/k$, for a centred cell in its conventional basis. The reduced superbase is $F_0 = (d, a, b, c)$, $d = -a - b - c$; in the reduced basis its four

vectors are the rows $(-1,-1,-1)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$. The reference's widened set at strain $\eta = e^{\epsilon}-1$ for the price band ϵ has members $F_1, \ldots, F_m$; each is a superbase of four integer vectors, written as rows in the reference's reduced basis, with a depth, the strain from $\mathbf{G}_R$ to the nearest metric that makes it obtuse. The new cell N is reduced in the same way, which gives B_N, $\mathbf{G}_N^{\mathrm{in}}$ and $\mathbf{G}_N = B_N^{\mathsf{T}} \mathbf{G}_N^{\mathrm{in}} B_N$. The price of an operator $\mathbf{P}$ is

$$\pi(\mathbf{P}) = \min_{s>0} D\big(\mathbf{G}_N^{\mathrm{in}},\, s\,\mathbf{P}^{\mathsf{T}} \mathbf{G}_R^{\mathrm{in}} \mathbf{P}\big), \tag{3}$$

with D the geodesic distance of equation (3) of the main text.

S10.2. Step 1: the member table

Ordered bases. A superbase $F = (v_0, v_1, v_2, v_3)$ has 48 labelled bases: leave out one of the four vectors, put the other three in an order, and give them a common sign $c = \pm 1$. Write B for the 3×3 integer matrix whose rows are the three vectors in order, without the sign; there are $4 \times 3! = 24$ such *ordered bases* per superbase, and the 48 labelled bases are cB. Every B is unimodular, since three vectors of a superbase form a basis of the lattice.

Their metrics. The column operator of the labelled basis cB is $\mathbf{W} = cB^{\mathsf{T}}$, and the metric it predicts for the new cell's reduced basis is

$$\mathbf{H} = \mathbf{W}^{\mathsf{T}} \mathbf{G}_R \mathbf{W} = c^2\, B\, \mathbf{G}_R\, B^{\mathsf{T}} = B\, \mathbf{G}_R\, B^{\mathsf{T}}, \tag{4}$$

which does not depend on the sign. A member therefore predicts 24 metrics, not 48. The member table holds, for every member k and ordered basis o, the matrix B_{ko}, its determinant $\det B_{ko} = \pm 1$, and $\mathbf{H}_{ko}$; it is computed once per reference and serves every new cell compared with it. In the conorm language of Section 2 of the main text, $\mathbf{H}_{ko}$ is the metric whose reduced superbase has the six conorms of F_k in the slot order that the ordering o induces, so the table is the member's six conorms under the 24 permutations of the four vectors.

S10.3. Step 2: the new cell, and why the input bases drop out

The new cell is reduced once. For $\mathbf{P} = B_R \mathbf{W} B_N^{-1}$,

$$\mathbf{P}^{\mathsf{T}} \mathbf{G}_R^{\mathrm{in}} \mathbf{P} = B_N^{-\mathsf{T}}\, \mathbf{W}^{\mathsf{T}} \big(B_R^{\mathsf{T}} \mathbf{G}_R^{\mathrm{in}} B_R \big) \mathbf{W}\, B_N^{-1} = B_N^{-\mathsf{T}}\, \mathbf{H}\, B_N^{-1}, \qquad \mathbf{G}_N^{\mathrm{in}} = B_N^{-\mathsf{T}}\, \mathbf{G}_N\, B_N^{-1}. \tag{5}$$

Both metrics in equation (3) are transformed by the same congruence $X \mapsto B_N^{-\mathsf{T}} X B_N^{-1}$, an isometry of D (main text, Section 3.4); equivalently, $(\mathbf{G}_N^{\mathrm{in}})^{-1} \mathbf{P}^{\mathsf{T}} \mathbf{G}_R^{\mathrm{in}} \mathbf{P} = B_N\, \mathbf{G}_N^{-1} \mathbf{H}\, B_N^{-1}$ is similar to $\mathbf{G}_N^{-1}\mathbf{H}$ and has the same eigenvalues. Hence

$$\pi(\mathbf{P}) = \min_{s>0} D(\mathbf{G}_N,\, s\,\mathbf{H}), \tag{6}$$

the comparison of the new cell's reduced metric with an entry of the member table. Neither input basis, nor the rational part of a centred reference's B_R, enters a price. The implementation forms $\mathbf{G}_R$ and $\mathbf{G}_N$ from the input cells exactly as written, so that the prices agree with the direct form to rounding.

S10.4. Step 3: the determinant rule

$\det \mathbf{P} = \det B_R\, c^3 \det B \,/\, \det B_N$, so the sign of $\det \mathbf{P}$ is $c \,\det B\, \operatorname{sgn}(\det B_R) \operatorname{sgn}(\det B_N)$. For a proper operator (main text, Section 4.2) the sign c is therefore fixed by the ordered basis,

$$c = \det B\ \operatorname{sgn}(\det B_R)\ \operatorname{sgn}(\det B_N), \tag{7}$$

and every ordered basis gives exactly one proper candidate: $24m$ candidates for m members, each with its own metric. For a reference whose widened set is the whole first layer, $m = 127$ and $24 \times 127 = 3048$, the proper half of $\mathcal{U}$ that the lookup of Remark A8 tries (Table 7 of the main text, step 3). When improper operators are asked for, the other sign is taken; when both are, both are listed, with the same price.

S10.5. Step 4: one scale for every candidate

Every $\mathbf{H}_{ko}$ has $\det \mathbf{H}_{ko} = (\det B_{ko})^2 \det \mathbf{G}_R = \det \mathbf{G}_R$. With l_1, l_2, l_3 the logarithms of the eigenvalues of $\mathbf{G}_N^{-1/2} \mathbf{H} \, \mathbf{G}_N^{-1/2}$, $D(\mathbf{G}_N, s\mathbf{H})^2 = \sum_i (l_i + \ln s)^2$, which is smallest at $\ln s = -\bar{l}$, $\bar{l} = \frac{1}{3}\sum_i l_i = \frac{1}{3}\ln(\det \mathbf{H} / \det \mathbf{G}_N)$. So the best scale

$$s = \Big(\frac{\det \mathbf{G}_N}{\det \mathbf{G}_R} \Big)^{1/3} \tag{8}$$

is the same for every candidate, and the price is $\pi = \big(\sum_i (l_i - \bar{l})^2 \big)^{1/2}$. The limit on the fitted change of edge lengths, $|\sqrt{s} - 1| \le 10\%$, is therefore one test for the pair of cells, not one per candidate; since the ratio of the primitive volumes is $V_N / V_R = s^{3/2}$, it is the volume check of step 2 of Table 7 of the main text, $(0.9)^3 < V_N / V_R < (1.1)^3$, up to the treatment of the boundary.

S10.6. Step 5: a necessary condition on ten dot products

The condition. Let $u_0, \ldots, u_3$ be the four vectors of the new cell's reduced superbase, with dot products $u_i \cdot u_j$ under $\mathbf{G}_N$, lengths $|u_i|$ under $\mathbf{G}_N$, and dot products $u_i^{\mathsf{T}} \mathbf{H} u_j$ under a predicted metric. If $\pi \le \epsilon$, then

$$\big|\, s\, u_i^{\mathsf{T}} \mathbf{H} u_j - u_i^{\mathsf{T}} \mathbf{G}_N u_j \,\big| \le \eta \, |u_i| \, |u_j| \qquad (0 \le i \le j \le 3), \qquad \eta = e^{\epsilon} - 1. \tag{9}$$

Proof. By step 4, $\pi = D(\mathbf{G}_N, s\mathbf{H})$. By Lemma A7(i) of the main text with $\mathbf{A} = \mathbf{G}_N$ and $\mathbf{B} = s\mathbf{H}$, $\| s\mathbf{H} - \mathbf{G}_N \|_{\mathbf{G}_N} \le e^{\pi} - 1 \le \eta$. By Proposition A4 of the main text with the metric $\mathbf{G}_N$ and the perturbation $\Delta\mathbf{G} = s\mathbf{H} - \mathbf{G}_N$, $|u^{\mathsf{T}} \Delta\mathbf{G} \, v| \le \eta |u| |v|$ for all vectors u, v, in particular for u_i, u_j. □

The condition is exact, not a first-order estimate, and it never discards an ordering within the band. It reads the new cell's six conorms $p_{ij} = -u_i \cdot u_j$ and four squared lengths against the member's conorms in the slot order of the ordering, scaled by s: each conorm may differ by at most $\eta |u_i| |u_j|$ and each squared length by $\eta |u_i|^2$. All ten bounds are computed once per new cell, and the predicted dot products $u_i^{\mathsf{T}} \mathbf{H}_{ko} u_j$ are

entries of the member table, so the test costs ten comparisons per ordering. To absorb rounding, the implementation widens each bound by a relative 10^{-12} and an absolute $10^{-9}\max|u_i \cdot u_j|$, which only lets more orderings through.

Why it is selective. A wrong ordering puts a large conorm of the member in a slot where the new cell has a small one, or the reverse, and conorms of a reduced cell differ by much more than η times the product of lengths unless the cell is close to a symmetry that exchanges them. In the cases examined here an ordering passes when it is the true one, when it differs from the true one by an exact or near symmetry of the lattice, or when the member lies at a depth below about the band; the last two are the alternatives that a band is meant to report.

S10.7. Step 6: the prices

For the orderings that pass, the eigenvalues of $\mathbf{G}_N^{-1/2}\mathbf{H}_{ko}\mathbf{G}_N^{-1/2}$ are computed in one batch (3×3 symmetric eigenvalue problems), and $\pi = \|l - \bar{l}\mathbf{1}\|$. Orderings with $\pi > \epsilon$ are dropped; with the scale fit switched off, $\bar{l}$ is replaced by 0 and s by 1 in steps 4 to 6.

S10.8. Step 7: the exact operators

For each ordering within the band the exact operator is built in rational arithmetic,

$$\mathbf{P} = B_R \left(c\, B_{ko}^{\mathsf{T}}\right) B_N^{-1}, \tag{10}$$

with the sign of step 3; then $(a_N\, b_N\, c_N) = (a_R\, b_R\, c_R)\mathbf{P}$, $\mathbf{G}_N^{\mathrm{in}} \approx s\mathbf{P}^{\mathsf{T}}\mathbf{G}_R^{\mathrm{in}}\mathbf{P}$ and $h_N = \mathbf{P}^{\mathsf{T}}h_R$ (main text, equation (5)). Each candidate carries its price, the scale s, the depth of its member and its determinant. Two different pairs (k, o) cannot give the same $\mathbf{P}$: three ordered vectors determine the superbase they belong to, and the members are distinct superbases; the implementation checks for repeats all the same. The candidates are listed by price, ties nearest the identity first, as in the direct form, and are

then grouped into indexing classes (step 5 of Table 7; main text, Section 5.3).

S10.9. Why the result is the direct form's

The direct form prices $\pi(B_R\, cB^{\mathsf{T}} B_N^{-1})$ for every member, ordered basis and sign. By step 2 each such price is $\min_s D(\mathbf{G}_N, s\mathbf{H}_{ko})$, by step 3 the sign only decides whether the operator is proper, by step 4 the minimising scale is the common s, and by step 5 an ordering discarded by the necessary condition has $\pi > \epsilon$. The workflow therefore returns exactly the operators of the direct form within the band, with the same prices up to rounding; the completeness of the candidate set (Lemma A7 and the completeness of $\mathcal{U}$) carries over unchanged. On the 780 cases of the benchmark the two return the same set of operators in every case, and on 24 cases with centred references (four new cells in random primitive bases for each of $C121$, $I222$, $R3$ on hexagonal axes, $F23$, $I4_1$ and $Fddd$) as well. The prices agree to 3×10^{-12}; the workflow gives exactly equal prices to operators related by a symmetry of the reference, since they share a predicted metric, where the direct form's products of rational and floating-point matrices differ in the last digits.

S10.10. Cost

Table S16. *The work of one reindexing, for a reference with m members, in the direct form and in the workflow, with the counts for the worked cell (Table S17) and, in parentheses, the medians and the totals over the 780 cases of the benchmark.*

	direct form	workflow
once per reference	–	member table: $24m$ ordered bases and metrics (384)
exact operators built	$48m$ (768; 238 224 in all)	those within the band (2; 5 797 in all)
dot-product conditions	–	$10 \times 24m$ comparisons (3 840; median 72 orderings)
eigenvalue problems	$48m$ (768)	orderings that pass (4; median 4, 5.9% of all)
time for the 780 new cells	about 27 s	about 1.3 s

The direct form builds and prices an exact rational operator for each of the $48m$ labelled bases. The workflow builds the member table once per reference, $24m$ integer

matrices and 3×3 products; for each new cell it computes two determinants, ten bounds and $24m$ sets of ten comparisons, solves an eigenvalue problem only for the orderings that pass, and builds exact operators only within the band (Table S16). On the benchmark the members number 3 at the median and 56 at most, the orderings 72 at the median and 1 344 at most; 5.9% of the orderings pass the condition (median 4 per case, at most 64), and the work of building exact operators falls from 238 224 to 5 797, which is also the number of operators returned. The 780 reindexings took about 27 s in the direct form and about 1.3 s in the workflow on the same machine, a factor of about 20; the reference was widened beforehand in both, and the times depend on the machine.

S10.11. The worked cell

Table S17. *The workflow on the worked cell: the reference* $30.02 \times 39.97 \times 50.03$ *Å,* 90.25, 92.05, 89.80° *at the default price band, and as new cell the same lattice in the basis* $(a, b, a + c)$ *with an added strain of* 0.3%. *Conorms in* $Å^2$ *in the slot order* da, db, dc, ab, ac, bc.

quantity	value
price band ϵ; strain $\eta = e^{\epsilon} - 1$	0.0740; 0.076807
reference: B_R, $\det B_R$	$((-1,0,1),(0,1,0),(0,0,1))$, -1
reference conorms	49.54, 1588.88, 2444.74, 4.19, 847.47, 4.54
widened set; layer	16 members (Table S18); first layer
new cell	30.0319, 40.0110, 57.4349 Å; 90.0718, 60.5600, 89.8328°
new cell: B_N, $\det B_N$	$((1,0,0),(0,0,-1),(0,-1,0))$, -1
new conorms	50.61, 2448.09, 1594.50, 847.80, 3.51, 2.88
step 1: ordered bases	$16 \times 24 = 384$, each with one proper sign (step 3)
step 4: common scale s; $\sqrt{s}$	1.001225; 1.000612, within 10%
step 5: bounds $\eta\|u_i\|\|u_j\|$, rows	$(314.4, 147.6, 282.2, 196.6)$, $(147.6, 69.3, 132.5, 92.3)$,
and columns d, a, b, c	$(282.2, 132.5, 253.4, 176.5)$, $(196.6, 92.3, 176.5, 123.0)$
step 5: orderings that pass	4 of 384 (Table S19)
step 6: within the band	2
step 7: operators	$((1,0,1),(0,1,0),(0,0,1))$, price 0.212%, the true one; $((-1,0,-1),(0,1,0),(0,0,-1))$, price 1.381%, the pseudo-twofold
direct form	768 exact operators built, the same two returned, prices equal to 7×10^{-16}

Table S18. *The 16 members of the worked cell's widened set at the default band: their depths (strain to the nearest metric that makes them obtuse) and their conorms (Å^2) under the reference metric, in each member's stored labelling. The members are those of Table S6 in another order and labelling; the conorms agree up to a permutation of the slots. The first six form the ring of Fig. 1 of the main text; the others lie at depths of* 3.8 *to* 5.1%.

member	depth (%)	conorms					
0	0.000	2444.7	1588.9	49.5	4.5	847.5	4.2
1	0.280	852.0	1593.4	45.0	−4.5	2449.3	8.7
2	0.494	8.7	2440.6	53.7	1593.1	−4.2	851.7
3	0.617	2458.0	−8.7	53.7	1602.1	843.3	4.2
4	0.777	847.5	−4.2	57.9	−4.5	2453.8	1606.3
5	0.777	2449.3	−8.7	62.5	4.5	842.9	1601.8
6	3.757	−49.5	2494.3	1638.4	53.7	897.0	−45.0
7	4.561	−62.5	2511.7	53.7	1664.2	905.4	−57.9
8	5.056	1584.7	950.7	2548.0	4.2	8.7	−53.7
9	5.078	1539.3	49.5	2494.3	905.4	62.5	−53.7
10	5.078	12.9	954.9	2543.8	−4.2	1588.9	−49.5
11	5.091	12.9	2556.7	942.0	−8.7	1593.4	−45.0
12	5.091	1548.4	897.0	45.0	57.9	2511.7	−53.7
13	5.111	−12.9	967.8	2556.7	1601.8	8.7	−62.5
14	5.111	−12.9	2569.6	954.9	1606.3	4.2	−57.9
15	5.111	1610.5	959.1	2565.5	−4.2	−8.7	−53.7

Table S19. *The four orderings of the worked cell that pass the necessary condition of step 5: the member, its superbase (rows in the reference's reduced basis), the vector left out and the order of the other three, the sign that makes the operator proper, the ordered basis with that sign, and the price. The first two are within the band.*

member	superbase (v_0, v_1, v_2, v_3)	ordering	sign	basis rows	price (%)
0	$(-1,-1,-1)$, $(0,0,1)$, $(0,1,0)$, $(1,0,0)$	omit v_0; v_3, v_1, v_2	-1	$(-1,0,0)$, $(0,0,-1)$, $(0,-1,0)$	0.212
4	$(-1,0,0)$, $(0,0,-1)$, $(0,1,0)$, $(1,-1,1)$	omit v_3; v_0, v_1, v_2	-1	$(1,0,0)$, $(0,0,1)$, $(0,-1,0)$	1.381
10	$(-2,0,-1)$, $(0,-1,0)$, $(1,0,0)$, $(1,1,1)$	omit v_3; v_2, v_0, v_1	-1	$(-1,0,0)$, $(2,0,1)$, $(0,1,0)$	10.194
14	$(-2,0,-1)$, $(0,1,0)$, $(1,-1,1)$, $(1,0,0)$	omit v_2; v_3, v_0, v_1	$+1$	$(1,0,0)$, $(-2,0,-1)$, $(0,1,0)$	10.208

Table S17 follows one reindexing through every step. The reference is the worked cell of the main text, widened at the default band: 16 members, the six of the ring and ten deeper ones (Table S18). The new cell is the same lattice in the basis (a, b, $a+c$), the other cell shape of Section 3.2 of the main text, with a strain of 0.3% added, so that its reduced cell is a different reduced cell of the lattice: its conorm ac of 3.5 Å^2 and bc of 2.9 Å^2 stand where the reference has 4.2 and 4.5 on ab and bc. Of the 384 ordered bases of the members, each with the sign that makes the operator proper,

four satisfy the ten inequalities (9) (Table S19); for each of the other 380 orderings at least one conorm, and at least one squared length, differs from the new cell's by more than its bound. Two of the four are within the band: the true operator, at 0.212%, the shape part of the strain that was added, and the true operator composed with the pseudo-twofold of the lattice, at 1.381%, from a member at depth 0.777% on the opposite side of the ring; as a self-map of the reference the pseudo-twofold costs 1.55% (main text, Table 9), and the added error moves that price. The other two, at 10.2%, pass the necessary condition but not the band; they are the cheapest of the reference's members outside it, and a band of 10.2% or more, with the reference widened for it, would return them. The direct form builds 768 exact operators to return the same two.

S10.12. Summary of the workflow

Table S20. *The reindexing workflow, with the result of each step that the implementation reports.*

step	action	reported
0	Widen the reference at $\eta = e^{\epsilon} - 1$ (main text, Table 7, step 1). Once per reference.	members, depths, layer
1	Member table: for every member the 24 ordered bases B, $\det B$, and $\mathbf{H} = B\mathbf{G}_R B^{\mathsf{T}}$. Once per reference.	$24m$ orderings
2	Reduce the new cell once; form $\mathbf{G}_N$ and $\mathbf{G}_R$ from the input cells.	B_N, conorms
3	Sign of each ordered basis from $\det B \operatorname{sgn} \det B_R \operatorname{sgn} \det B_N$.	number with the selected sign
4	Common scale $s = (\det \mathbf{G}_N / \det \mathbf{G}_R)^{1/3}$; test $\lvert\sqrt{s} - 1\rvert$ against the scale limit once.	s, $\sqrt{s}$
5	Bounds $\eta \lvert u_i \rvert \lvert u_j \rvert$ for the new reduced superbase; keep the orderings with $\lvert s\, u_i^{\mathsf{T}} \mathbf{H} u_j - u_i \cdot u_j \rvert$ within them for all $i \le j$.	bounds, number passing
6	Price the survivors in one batch: eigenvalues of $\mathbf{G}_N^{-1/2} \mathbf{H} \mathbf{G}_N^{-1/2}$, $\pi = \lVert l - \bar{l}\mathbf{1} \rVert$; keep $\pi \le \epsilon$.	prices, number within the band
7	Exact operator $\mathbf{P} = B_R\, cB^{\mathsf{T}} B_N^{-1}$ for each; order by price, ties nearest the identity.	candidates
8	Group into indexing classes (main text, Section 5.3).	classes

Table S20 collects the steps. Steps 0 and 1 are paid once per reference and serve every new cell, which is the case of serial crystallography and of a series of data sets against one reference; steps 2 to 7 cost one reduction, a few determinants, ten comparisons

per ordering, a handful of 3×3 eigenvalue problems and the exact operators that are returned.

S11. The Niggli window and the way to the Niggli cell: the finite proof

This section gives the computation behind Proposition A12 and Corollary A13 of the main text (claims z68 of Section S7, `sellingproofs/niggli_window.py`). Throughout, a zero pattern Z is fixed and vectors are written in the basis (a, b, c) of a superbase $F_0 = (d, a, b, c)$; the metrics for which F_0 is obtuse with pattern Z are $\sum_k p_k E_k$ with $p_Z = 0$ and $p_k > 0$ off Z, so that every squared length and every scalar product is a linear form in the conorms p, with integer coefficients.

Step 1: the vectors shortest in their classes. By Lemma A1 of the main text, a vector with coefficients $s + 2k$ relative to F_0 ($s \in \{0,1\}^4$, $s_0 = k_0 = 0$) is shortest in its parity class exactly when $\gamma_{ij} = \Delta k(\Delta k + \Delta s) = 0$ on every slot off Z. The condition does not involve the values of p, and along the connected graph of non-zero slots it gives $|k_i| \leq 3$, so the set $V(Z)$ of such vectors is listed exactly from the window $[-7, 7]^3$. It has 14, 16, 18, 20 and 26 vectors for the truncated octahedron, the hexarhombic dodecahedron, the rhombic dodecahedron, the hexagonal prism and the cuboid (Table S21); on three seeded metrics of every pattern the class minima found by enumeration are exactly $V(Z)$.

Step 2: the Minkowski-reduced bases. A Niggli cell is Minkowski reduced (Gruber, 1973), and each vector of a Minkowski-reduced basis is shortest in its class (Section S12, proof of Proposition A12), so every Niggli cell of every metric with pattern Z is an ordered basis of $V(Z)$ that satisfies the ten conditions

$$|b|^2 - |a|^2 \geq 0, \quad |c|^2 - |b|^2 \geq 0, \quad |b \pm a|^2 - |b|^2 \geq 0, \quad |c + xa + yb|^2 - |c|^2 \geq 0 \ \ (x, y \in \{-1, 0, 1\},\ (x, y) \neq 0)$$

at its own conorms. Each condition is a row A_r with $A_r p \geq 0$. An ordered basis is *retained* for Z when a linear program finds a point with $Ap \geq 0$, $p_Z = 0$ and $p_k \geq 1$

off Z (the conditions are homogeneous, so this is the relative interior of the face). A basis the program rejects is excluded only when an exact certificate is found: rational $y \geq 0$ with $y^{\mathsf{T}}A \leq 0$ on the free conorms and $y^{\mathsf{T}}A \neq 0$, verified in exact arithmetic, so that $y^{\mathsf{T}}Ap < 0$ for every p in the relative interior and the conditions cannot all hold. A basis the program accepts is retained without further proof, which can only enlarge the set checked. Of the 174 912 ordered bases of $V(Z)$ over the 38 patterns, 8 256 are retained and the other 166 656 are excluded, every one with a verified certificate.

Step 3: the window. For each pattern, every pair (B, B') of retained bases gives $B^{-1}B'$; all 3 849 216 have entries in $\{-1, 0, 1\}$. The pairs are not required to be reduced at a common p, which is what part (ii) of the proposition uses: two measurements near one lattice have Niggli cells that are each retained for its pattern, at different conorms.

Step 4: the way from a reduced cell. A superbase of vectors of $V(Z)$ is obtuse for the metrics with pattern Z exactly when its seven partial sums lie in $V(Z)$ (Section S12, step (3) of the enumeration); as a cross-check, these are also exactly the superbases for which a linear program finds the six conditions $v_i \cdot v_j \leq 0$ satisfiable on the face. They number 1, 2, 3, 6 and 16 up to sign, as in Lemma A2. For every one, each of its 24 ordered bases (u_1, u_2, u_3) and each retained basis B, the matrix $(u_1, u_2, u_3)^{-1}B$ has entries in $\{-1, 0, 1\}$: 6 856 704 matrices, 1 344 of them distinct. These 1 344 are exactly the unimodular matrices whose columns are three of the 14 vectors $\pm(1,0,0)$, $\pm(0,1,0)$, $\pm(0,0,1)$, $\pm(1,1,0)$, $\pm(1,0,1)$, $\pm(0,1,1)$, $\pm(1,1,1)$, the signed partial sums of the superbase in the basis (u_1, u_2, u_3); 672 of them have determinant $+1$. So the Niggli vectors are three of the partial sums of every obtuse superbase, and the list does not depend on the lattice. Requiring the superbase to be obtuse matters: among all superbases with four vectors in $V(Z)$, some reach a retained basis only through an entry of 2.

Step 5: the radius. For part (ii), η_{gap} is the class gap $(g-1)/(g+1)$ of the main text,

computed by enumerating the classes, and η_{adm} is the smallest strain, to first order, at which an excluded basis of $V(Z)$ satisfies the ten conditions, a convex problem per basis solved numerically. In every example ρ is set by the class gap (Table S21); two measurements of each example lattice within 0.9ρ, the second in a random basis, had Niggli cells within $\{-1, 0, 1\}$ of each other in all 280 trials, and the library's Selling and Niggli reductions (the latter after Grosse-Kunstleve *et al.*, 2004), run on 380 seeded cells on and slightly off every pattern, gave Niggli cells that satisfy the ten conditions and lie within $\{-1, 0, 1\}$ of all 24 ordered bases of the reduced superbase.

Step 6: the Niggli cell from the reduced cell. The library reads the Niggli cell off the Selling-reduced superbase (`niggli_from_selling` in `agentsg.cell.strain_closure`). It forms the 672 right-handed ordered bases of the 14 signed partial sums, computes their six scalars $A = a \cdot a$, B, C, $\xi = 2b \cdot c$, $\eta = 2a \cdot c$, $\zeta = 2a \cdot b$ in one batch, and keeps the basis that satisfies Niggli's conditions (Křivý & Gruber, 1976) in the form of the termination tests of Grosse-Kunstleve *et al.* (2004), with the same relative tolerance. When the tolerance lets several bases pass, the one with the smallest scalars is returned. On 15 200 seeded cells, on, near and on both sides of every zero pattern, with tied conorms, of both Niggli types and in random bases, it returned the cell of the library's Niggli reduction in every case, in about 3 ms per cell (claim z68); a smaller sample is the library's own test (claim z63). In that sample 516 of the 672 matrices occurred. This suggests, but does not show, that a shorter list would suffice: the list is proved complete, not minimal.

Table S21. *The finite proof of Proposition A12 and Corollary A13 by Voronoi type (top), and the radius ρ of Proposition A12(ii) for example lattices (bottom; strain, %). $|V|$: vectors shortest in their classes; bases: ordered bases of V; retained: Minkowski reduced somewhere on the pattern, the rest excluded with exact certificates; superbases: obtuse, up to sign; pairs: the matrices checked for Proposition A12(i) and Corollary A13, all with entries in $\{-1, 0, 1\}$.*

Voronoi type	patterns	$\lvert V\rvert$	bases	retained	superbases	pairs (A12)	pairs (A13)
truncated octahedron	1	14	1 344	768	1	589 824	147 456
hexarhombic dodecahedron	6	16	11 520	2 688	2	1 204 224	1 032 192
rhombic dodecahedron	3	18	7 488	2 304	3	1 769 472	1 327 104
hexagonal prism	12	20	43 200	1 728	6	248 832	1 990 656
cuboid	16	26	111 360	768	16	36 864	2 359 296
all	38		174 912	8 256		3 849 216	6 856 704

lattice	zeros	η_{gap}	η_{adm}	ρ
mP, $30 \times 40 \times 50$ Å, $\beta = 100°$	2	10.4	46.9	10.4
oC, primitive cell $25 \times 50 \times 25$ Å, $\beta = 106.3°$	2	9.4	33.3	9.4
hP, $40 \times 40 \times 60$ Å	2	23.5	42.6	23.5
oP, $30 \times 40 \times 50$ Å	3	30.5	39.1	30.5
tP, $40 \times 40 \times 60$ Å	3	38.1	42.6	38.1
cP, 40 Å	3	50.0	57.7	50.0
aP, $30 \times 40 \times 50$ Å, $85, 95, 100°$	0	1.9	28.9	1.9

S12. Statements and proofs

This section gives in full the statements summarised in Appendix A of the main text, with their proofs, under the same numbers. Each has a computational check, a machine-checked proof or both (Section S7); Table 6 of the main text records which.

Terms. Two lattice vectors are in the same *parity class* when their difference is twice a lattice vector; the classes are the eight cosets of 2Λ in Λ, one of them 2Λ itself, and the other seven are used here. In a basis v_1, v_2, v_3 every lattice vector is $x = n_1v_1 + n_2v_2 + n_3v_3$ with integer coefficients n_i, and $x - y$ is twice a lattice vector exactly when every coefficient of $x - y$ is even. So a vector's class is read from the parities of its coefficients, the pattern $(n_1, n_2, n_3) \bmod 2$, with $(0, 0, 0)$ for 2Λ. The pattern is a label that depends on the basis: a unimodular change of basis maps the coefficients by an integer matrix whose inverse is also integer, which keeps even coefficients even, so it permutes the labels and leaves the classes unchanged. For a superbase (v_0, v_1, v_2, v_3) with $v_0 = -v_1 - v_2 - v_3$ we write a lattice vector in the basis

v_1, v_2, v_3, $x = \sum_{i=1}^{3} c_i v_i$, and set $c_0 = 0$. Where a formula treats the four vectors alike it is written as $\sum_{i=0}^{3} c_i v_i$; since the four vectors sum to zero, adding the same integer t to all four c_i gives the same vector, and such formulas depend only on the differences $c_i - c_j$. An odd t complements the parity pattern, so a pattern $s \in \{0,1\}^4$ and $1 - s$ name the same class. A *partial sum* of a superbase is the sum of some of its four vectors. Since the four sum to zero, the sum over a subset is minus the sum over its complement, and a vector and its negative lie in one parity class; so, up to sign, the partial sums are the seven sums over non-empty subsets of $\{v_1, v_2, v_3\}$, one in each class other than 2Λ. With $c_0 = 0$ their patterns are $s = (0, s_1, s_2, s_3)$ with $(s_1, s_2, s_3) \neq (0,0,0)$. A superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class, that is, a Voronoi vector of the lattice (Voronoi, 1908; Conway & Sloane, 1992; Kurlin, 2022). A Voronoi vector v is one whose half, $v/2$, lies on the boundary of the Voronoi (Wigner–Seitz) cell of the origin; it is *strict* when $\pm v$ are the only shortest vectors of its class, and then the plane bisecting v carries a face of the cell. At a zero conorm the class gains a second pair of shortest vectors: with $a \cdot b = 0$, $|a - b|^2 = |a + b|^2$, so $a + b$ and $a - b$ are both shortest in their class, neither is strict, and the cell has no face normal to either; the reduced superbase, whose partial sum in that class is $a + b$, and the one reached by the flip at ab, whose partial sum there is $b - a$, are then both obtuse. The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase F in class c is the squared length of its partial sum in that class minus the squared length μ_c of the shortest vector of the class. A partial sum is written $w = \sum_{i=0}^{3} s_i v_i$ with $s \in \{0,1\}^4$; a vector of the same class has coefficients $s + 2k$ with k integer. Every lattice vector $x = \sum_{i=0}^{3} c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij} \, (c_i - c_j)^2, \tag{11}$$

whatever the signs of the conorms, because the four vectors sum to zero.

Lemma A1 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ a partial

sum and w' a vector of its parity class with coefficients $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + \Delta s)$, $\Delta k = k_i - k_j$, $\Delta s = s_i - s_j$; each γ_{ij} is a non-negative integer, zero exactly when $\Delta k \in \{0, -\Delta s\}$. *Proof.* Equation (11), $(\Delta s + 2\Delta k)^2 - \Delta s^2 = 4\Delta k(\Delta k + \Delta s)$, and $t(t + u) \geq 0$ for integers t and $u \in \{-1, 0, 1\}$. □

Lemma A2 (window). Let F_0 and F be obtuse superbases of the same lattice. Then every vector of F has superbase coefficients $s_i + 2k_i$ relative to F_0, with $s \in \{0, 1\}^4$, $s_0 = k_0 = 0$ and $|k_i| \leq 3$, so that $c_0 = 0$ (in the basis (a, b, c) the coefficients then lie in $[-6, 7]$; the machine-checked version uses the window $[-7, 7]$). *Proof.* The non-zero conorms of F_0 connect its four vectors, since otherwise a partial sum would have length zero by equation (11). Every partial sum of F is a shortest vector of its parity class, as is the partial sum of F_0 in that class, so their excess vanishes; by Lemma A1 the excess is $4\sum p^0_{ij}\gamma_{ij}$ with every term non-negative, so $\gamma_{ij} = 0$ on every non-zero slot, i.e. $|k_i - k_j| \leq 1$ along a connected graph on four vertices, and with $k_0 = 0$ every $|k_i| \leq 3$. □

The enumeration. Lemma A2 reduces the completeness of $\mathcal{U}$ to a finite computation, carried out as follows. (1) *Zero patterns.* A zero pattern is a set of at most three of the six slots whose complement connects the four vectors; there are 38. A conorm vector with a given pattern has $p_{ij} > 0$ exactly off the pattern. (2) *Candidate vectors.* For each pattern, list the integer vectors x in the window of Lemma A2 whose coefficients $s + 2k$ relative to F_0 have $\gamma_{ij} = 0$ on every slot off the pattern: by Lemma A1 these are exactly the vectors that are shortest in their parity class for every metric with that pattern. (3) *Candidate superbases.* Form every triple (u_1, u_2, u_3) of listed vectors with $\det(u_1, u_2, u_3) = \pm 1$ (unimodularity) whose seven partial sums are all listed; with $u_0 = -u_1 - u_2 - u_3$ these are the superbases whose partial sums are all shortest, that is, the obtuse superbases, for every metric with that pattern. Because the condition is the vanishing of integers on the slots off the pattern, it holds for one metric with

the pattern exactly when it holds for all, so one generic metric per pattern stands for every lattice with it. (4) *Duplicates.* Two triples give the same superbase when they are relabellings of each other; superbases are kept as unordered sets of four vectors up to a common sign. (5) *The set $\mathcal{U}$.* Over all 38 patterns, the matrices whose columns are the three vectors of a superbase, in each of its 48 labellings, form $\mathcal{U}$; that every matrix between two obtuse superbases of any lattice is among them follows from Lemma A2 and steps (2)–(3). The enumeration gives 1, 2, 3, 6 and 16 superbases for no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector and three zeros (both placements), independently of Kurlin's lists. In the basis of F_0, over every labelling of each member, their entries are in $\{-1, 0, 1\}$ when the zero conorms avoid d, and up to 3 otherwise (Table 4 of the main text): the conorms $(0, 0, 3, 2, 0, 5)$ have the obtuse basis $(-3a - 2b - c,\ a,\ a + b)$. Every member lies in $\mathcal{U}$ in each of its 48 labellings, and every member has a labelling with entries in $[-2, 2]$. Relabelling F_0 once for all members, the largest entry is 1 on every pattern but the path of three zeros, where it is 2; relabelling both F_0 and each member, every entry is in $\{-1, 0, 1\}$. The machine-checked version (Section S7) runs steps (2)–(3) over the window $[-7, 7]^3$ for every subset of slots and checks that every superbase found is in the deposited table.

The 127 superbases by hand. The union of the closures of all lattices whose reduced cell is F_0, over the 38 patterns, is the first layer $\mathcal{U}F_0$. A pattern contributes the superbases of its closure not already contributed by the patterns it contains: the reduced cell itself (no zero), one for each of the 6 single zeros (the boundary transformation at that zero), 3 for each of the 12 pairs of zeros sharing a vector (the hexagonal ring of six closes with three new members), 3 for each of the 4 triangles and 6 for each of the 12 paths of three zeros (the cuboid's sixteen superbases less those of its sub-patterns), and none for the 3 pairs of zeros on disjoint pairs, whose three superbases are the reduced cell and the two single steps. So $|\mathcal{U}F_0| = 1 + 6 + 36 + 12 + 72 = 127$, and

$|\mathcal{U}| = 48 \times 127$; the enumeration reproduces each contribution.

Lemma A3 (generation). Every matrix of the closure, and every symmetry, is a word of boundary transformations (Selling steps at zero conorms) followed by a relabelling. *Proof.* Let F and F' be obtuse superbases of a lattice with metric $\mathbf{G}$. Add a small positive amount, in the basis of F, to each conorm of F that vanishes under $\mathbf{G}$; this gives a metric $\mathbf{G}'$ for which F is strictly obtuse and hence the only obtuse superbase, and, the lattice having finitely many obtuse superbases, the amount can be chosen so small that every conorm that is positive under $\mathbf{G}$ for any of them stays positive under $\mathbf{G}'$. Selling reduction under $\mathbf{G}'$ started from F' takes steps only at negative conorms. The first such conorm is zero under $\mathbf{G}$, so the step carries F' to another obtuse superbase of $\mathbf{G}$, and the same holds at every later step. Reduction terminates (Selling, 1874; Conway & Sloane, 1992), at the only obtuse superbase of $\mathbf{G}'$, which is F. So F' is joined to F by boundary transformations of $\mathbf{G}$, and a relabelling matches the labellings. A symmetry maps F_0 to an obtuse superbase of the lattice; as a matrix it is that word followed by a relabelling. □ Kurlin (2022) lists the superbases of each Voronoi type, and the enumeration of Lemma A2 reproduces the counts.

Proposition A4 (the strain norm bounds every dot product). If $\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F \leq \eta$, then $|u^{\mathsf{T}}\Delta\mathbf{G}\,v| \leq \eta\,|u|\,|v|$ for all vectors u, v, lengths in $\mathbf{G}$; in particular every conorm of every superbase changes by at most $\eta|u_i||u_j|$. *Proof.* With $X = \mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}$, $u^{\mathsf{T}}\Delta\mathbf{G}\,v = (\mathbf{G}^{1/2}u)^{\mathsf{T}}X(\mathbf{G}^{1/2}v)$, at most $\|X\|_2|u||v| \leq \|X\|_F|u||v|$. □

Definition and enumeration A5 (sorts as closed sets of equalities). Fix a zero pattern Z, let O_Z be the matrices of its closure and, for $\mathbf{W} \in O_Z$, let $\mathbf{A}_\mathbf{W}$ be its linear action on conorm vectors (Section S3), which on the face $\{p \geq 0,\ p_Z = 0\}$ is a permutation of the slots. For a subspace V of the face put $\mathrm{Sym}(V) = \{\mathbf{W} \in O_Z : \mathbf{A}_\mathbf{W} = \mathrm{id} \text{ on } V\}$ and $\mathrm{Fix}(H) = \{p : \mathbf{A}_\mathbf{W}p = p\ \forall\,\mathbf{W} \in H\}$. A lattice with conorm vector p on the face has point group $\mathrm{Sym}(\{p\})$, every symmetry being in O_Z by Lemma A3. We take

as the definition of a sort a maximal set of lattices with the same kind of closure and the same symmetry operators; then the sorts of kind Z are the generic points of the closed subspaces $V = \mathrm{Fix}(\mathrm{Sym}(V))$ that meet the interior of the face, modulo the matrices that map the face onto itself. This is the classical construction, the fixed subspace of a point group in the space of forms (Delone *et al.*, 1975), written in conorm coordinates; SELLA uses the same subspaces. Sym and Fix are a Galois connection, so the closed subspaces are the intersections of the finitely many subspaces $\mathrm{Fix}(\mathbf{W})$; since $\mathbf{A}_\mathbf{W}$ permutes the slots on the face, each is a set of equalities among conorms. Enumerating the intersections gives 8, 5, 3, 5 and 3 closed subspaces for no zero, one zero, two sharing a vector, two disjoint and three zeros, with symmetry orders $2, 4, 4, 8, 8, 12, 16, 48$; $2, 4, 4, 8, 16$; $4, 8, 24$; $2, 4, 8, 12, 48$; $8, 16, 48$. A generic point of a closed subspace, one in the interior of the face and in none of the finitely many smaller closed subspaces it contains, has exactly the symmetries $\mathrm{Sym}(V)$; points on its boundary belong to another zero pattern. Matched pattern by pattern, by computation, these are Delone's 24 sorts (Table S10).

Lemma A6 (the widened set is finite). If F is within strain $\eta < 1$ of obtuse for $\mathbf{G}$, every vector u of F satisfies $|u|^2_\mathbf{G} \le \frac{1+\eta}{1-\eta}\,\mu_c(\mathbf{G})$, with μ_c the squared length of the shortest vector of the parity class of u, which is that of F_0's partial sum in the class. *Proof.* Let $\mathbf{G}'$ within η make F obtuse. By Proposition A4 with $u = v$, $(1-\eta)|u|^2_\mathbf{G} \le |u|^2_{\mathbf{G}'} \le (1+\eta)|u|^2_\mathbf{G}$ for every lattice vector. The vectors of F are shortest in their classes for $\mathbf{G}'$, and those of F_0 for $\mathbf{G}$, so $|u|^2_{\mathbf{G}'} \le |v^0_c|^2_{\mathbf{G}'} \le (1+\eta)\mu_c(\mathbf{G})$ and $|u|^2_\mathbf{G} \le |u|^2_{\mathbf{G}'}/(1-\eta)$. □ The lattice vectors in these seven balls are finitely many and computable from $\mathbf{G}$, so the candidates for $\mathcal{S}_\eta(\mathbf{G})$ are enumerated exactly.

Lemma A7 (widening one side suffices). (i) If $D(\mathbf{A}, \mathbf{B}) \le \epsilon$, then $\|\mathbf{B}-\mathbf{A}\|_\mathbf{A} \le e^\epsilon - 1$. (ii) Let $\mathbf{P}$ be an operator from a reference with metric $\mathbf{G}_R$ to a new cell with metric $\mathbf{G}_N$, of price at most ϵ, and let F be obtuse for $\mathbf{G}_N$. Then $\mathbf{P}F \in \mathcal{S}_{e^\epsilon-1}(\mathbf{G}_R)$. *Proof.*

(i) The eigenvalues of $\mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}$ are e^{l_i} with $\sum l_i^2 \le \epsilon^2$, and $|e^l-1| \le |l|(e^a-1)/a$ for $|l| \le a$, because $(e^x-1)/x$ increases; with $a = \max|l_i| \le \epsilon$, $\|\mathbf{B}-\mathbf{A}\|_{\mathbf{A}} = (\sum(e^{l_i}-1)^2)^{1/2} \le \epsilon\,(e^\epsilon-1)/\epsilon$. (ii) The price is $D(\mathbf{G}_N, s\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P})$ for the fitted scale s. Carried back by $\mathbf{P}$, an isometry of both measures, $\mathbf{P}F$ is obtuse for $\mathbf{M} = \mathbf{P}^{-\mathsf{T}}\mathbf{G}_N\mathbf{P}^{-1}$, and $D(s\mathbf{G}_R, \mathbf{M}) \le \epsilon$, so by (i) $\mathbf{M}$ is within strain $e^\epsilon-1$ of $s\mathbf{G}_R$; the strain and the widened set are unchanged by the scale.□ For the price band of 7.4% the reference is widened at 7.7%.

Remark A8 (layers, and the lookup of $\mathcal{U}$). Two superbases are *adjacent* when some positive definite metric makes both obtuse; a superbase adjacent to F_0 is $\mathbf{W}F_0$ with $\mathbf{W} \in \mathcal{U}$, since both are superbases of that metric's closure. Let $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$, and define $r_k(\mathbf{G})$ as the smallest strain from $\mathbf{G}$ to a metric for which a superbase outside $\mathcal{N}_k$ is obtuse; so if $\eta < r_k(\mathbf{G})$, $\mathcal{S}_\eta(\mathbf{G})$ is the set of members of $\mathcal{N}_k$ within strain η of obtuse. The content of this definition is the description of $\mathcal{N}_1$, which rests on the completeness of $\mathcal{U}$. *The lookup, a corollary of Lemma A7 and the completeness of $\mathcal{U}$.* Let $\mathbf{P}$ be an operator of price at most ϵ from a reference with reduced basis B_R to a new cell with reduced basis B_N, and let the reference's widened set at $e^\epsilon-1$ lie in $\mathcal{N}_1$, as it does when $e^\epsilon-1 < r_1$. By Lemma A7 the new cell's reduced superbase, carried back by $\mathbf{P}$, lies in $\mathcal{S}_{e^\epsilon-1}(\mathbf{G}_R) \subseteq \mathcal{N}_1$, so it is $\mathbf{W}F_0$ for some $\mathbf{W} \in \mathcal{U}$ in some labelling, and $\mathbf{P} = B_R\mathbf{W}\mathbf{Q}B_N^{-1}$ with $\mathbf{Q}$ a relabelling. Since $\mathcal{U}$ is closed under relabelling, $\mathbf{W}\mathbf{Q} \in \mathcal{U}$: trying $B_R\mathbf{W}B_N^{-1}$ for the 6096 column operators $\mathbf{W}$, or for the 3048 with $\det\mathbf{W} = \det B_R \det B_N$ when $\mathbf{P}$ is proper, finds $\mathbf{P}$. □

The radii are computed from the balls of Lemma A6 at a strain of $\frac{1}{4}$, with the depths from a floating-point convex solver, and those of Table 8 of the main text are certified in exact arithmetic by a dual bound and a feasible point for each depth (Section S7). No lower bound on r_1 in terms of the cell's geometry is proved; for the cells of Table 8 of the main text below the cap it is of the order of the ratio of the shortest to the

longest edge.

Proposition A9 (the price of a symmetry). Let g be an integer matrix of finite order n and $V_g = \{\mathbf{G} : g^\mathsf{T}\mathbf{G}g = \mathbf{G}\}$ the metrics for which g is a symmetry. (i) $D(\mathbf{G}, g^\mathsf{T}\mathbf{G}g) \le 2D(\mathbf{G}, V_g)$, with equality when g is a twofold. (ii) If g is a twofold along a lattice row of obliquity δ in $\mathbf{G}$ (the angle between the row and the normal to the lattice plane that g negates), then $D(\mathbf{G}, g^\mathsf{T}\mathbf{G}g) = 2\sqrt{2}\,\mathrm{artanh}(\sin\delta)$. (iii) For a metric $\mathbf{G}^* \in V_g$ and the strain at $\mathbf{G}^*$, $\|g^\mathsf{T}\mathbf{G}g - \mathbf{G}\|_{\mathbf{G}^*}$ lies between $\rho_{\min}$ and $\rho_{\max}$ times the distance from $\mathbf{G}$ to V_g in that norm, with $\rho = |e^{i\theta} - 1|$ over the rotation angles $\theta \neq 0$ of $X \mapsto g^\mathsf{T}Xg$: $\rho = 2$ for a twofold, $\sqrt{3}$ for a threefold, $\{\sqrt{2}, 2\}$ for a fourfold and $\{1, \sqrt{3}\}$ for a sixfold. The scale fit leaves these prices unchanged, since $\det g^\mathsf{T}\mathbf{G}g = \det\mathbf{G}$. *Proof.* (i) $\mathbf{G} \mapsto g^\mathsf{T}\mathbf{G}g$ is an isometry of D. For any $\mathbf{G}' \in V_g$, $D(\mathbf{G}, g^\mathsf{T}\mathbf{G}g) \le D(\mathbf{G}, \mathbf{G}') + D(\mathbf{G}', g^\mathsf{T}\mathbf{G}g) = 2D(\mathbf{G}, \mathbf{G}')$. For a twofold, the geodesic midpoint m of $\mathbf{G}$ and $g^\mathsf{T}\mathbf{G}g$ is unique, the space of positive definite matrices with this distance being complete and of non-positive curvature (Bhatia, 2007, Chapter 6), and g exchanges the two ends, so $m \in V_g$ and $D(\mathbf{G}, V_g) \le \frac{1}{2}D(\mathbf{G}, g^\mathsf{T}\mathbf{G}g)$. (ii) In Cartesian coordinates for $\mathbf{G}$, g becomes $T = \mathbf{G}^{1/2}g\mathbf{G}^{-1/2}$, an involution fixing the line ℓ of the row and negating the plane Π; it is $2\mathbf{E} - I$ with $\mathbf{E}$ the projection onto ℓ along Π, and the angle between ℓ and the normal of Π is δ. The price is $\|\ln(T^\mathsf{T}T)\|_F$. $T^\mathsf{T}T$ is the identity on the line of Π normal to ℓ, and on the plane spanned by ℓ and the normal of Π it has eigenvalues $(1 \pm \sin\delta)/(1 \mp \sin\delta)$, whose logarithms are $\pm 2\,\mathrm{artanh}(\sin\delta)$. (iii) $\sigma : X \mapsto g^\mathsf{T}Xg$ is an isometry of the inner product $\mathrm{tr}(\mathbf{G}^{*-1}X\mathbf{G}^{*-1}Y)$ of order n, so the space of symmetric matrices splits orthogonally into its fixed space, which is $V_g - \mathbf{G}^*$, and planes and lines on which it rotates by $\theta = 2\pi k/n$. With $\mathbf{G} - \mathbf{G}^* = X_0 + X_\perp$, X_0 fixed, the distance to V_g is $\|X_\perp\|$ and $g^\mathsf{T}\mathbf{G}g - \mathbf{G} = (\sigma - 1)X_\perp$. The rotation angles are those of the products $\lambda_i\lambda_j$, $i \le j$, of the eigenvalues of g, squares included; the squares give the angle π of a fourfold and $2\pi/3$ of a sixfold. □ Part (iii) is exact for

the strain at $\mathbf{G}^*$; for the price it holds to first order in the distance to V_g. No exact relation to Le Page's obliquity holds for orders 3, 4 and 6: at Le Page's boundary the fourfolds of a body-centred cubic lattice cost a median 15.0%, more than its twofolds (Table S11). The metrics of a sort to which g belongs lie inside V_g (read in the basis in which the sort's distance is attained), so, to first order in the error, where the strain of a sort distance and the geodesic distance agree, a twofold's price is at most twice the distance to that sort, with equality when V_g is the sort's subspace, as for the twofold of a primitive monoclinic lattice.

Proposition A10 (candidates and cosets). Let $\mathbf{P}_0$ be the true operator from a reference R to a new cell N, of price p_0; let H_ϵ be the proper unimodular h with $\pi(h) = D(\mathbf{G}_R, h^{\mathsf{T}}\mathbf{G}_R h) \leq \epsilon$, a finite set, since $\pi(h) \leq \epsilon$ bounds the eigenvalues of $\mathbf{G}_R^{-1}h^{\mathsf{T}}\mathbf{G}_R h$ by e^ϵ, so that every column of h has $|he_i|^2 \leq e^\epsilon |e_i|^2$; and let C_ϵ be the candidates, the proper unimodular operators of price at most ϵ. Then (i) $\pi(h_1 h_2) \leq \pi(h_1) + \pi(h_2)$; (ii) if $p_0 \leq \epsilon$, $H_{\epsilon - p_0}\mathbf{P}_0 \subseteq C_\epsilon \subseteq H_{\epsilon + p_0}\mathbf{P}_0$; (iii) if no self-map of R has price in $(\epsilon, 2\epsilon]$, H_ϵ is a group; (iv) if none has price in $(\epsilon - p_0, \epsilon + p_0]$, then $C_\epsilon = H_\epsilon \mathbf{P}_0$; (v) if $\mathbf{G}_0$ is any metric at distance ν from $\mathbf{G}_R$, after normalisation to unit determinant, and $\mathrm{Aut}(\mathbf{G}_0)$ its point group, then $\mathrm{Aut}(\mathbf{G}_0) \subseteq H^{\pm}_{2\nu}(\mathbf{G}_R)$ and $H^{\pm}_\epsilon(\mathbf{G}_R) \subseteq H^{\pm}_{\epsilon+2\nu}(\mathbf{G}_0)$, where $H^{\pm}$ admits improper self-maps; in particular, if $\mathbf{G}_0$ has no self-map outside its point group of price at most $\epsilon + 2\nu$, and $\epsilon \geq 2\nu$, then $H^{\pm}_\epsilon(\mathbf{G}_R) = \mathrm{Aut}(\mathbf{G}_0)$. *Proof.* The price is the geodesic distance between metrics normalised to unit determinant, and unimodular matrices act on those by isometries. (i)

$$D(\mathbf{G}_R, (h_1h_2)^{\mathsf{T}}\mathbf{G}_R h_1 h_2) \leq D(\mathbf{G}_R, h_2^{\mathsf{T}}\mathbf{G}_R h_2) + D(h_2^{\mathsf{T}}\mathbf{G}_R h_2, h_2^{\mathsf{T}}h_1^{\mathsf{T}}\mathbf{G}_R h_1 h_2) = \pi(h_2) + \pi(h_1).$$

(ii) By the triangle inequality the price of $h\mathbf{P}_0$ differs from $D(\mathbf{P}_0^{\mathsf{T}}\mathbf{G}_R\mathbf{P}_0, \mathbf{P}_0^{\mathsf{T}}h^{\mathsf{T}}\mathbf{G}_R h\mathbf{P}_0) =$

$\pi(h)$ by at most p_0, and every unimodular $\mathbf{P}$ is $h\mathbf{P}_0$ with $h = \mathbf{P}\mathbf{P}_0^{-1}$. (iii) By (i) a product of two elements of H_ϵ has price at most 2ϵ, hence at most ϵ; a finite set of matrices closed under products is a group. (iv) The two sets of (ii) then coincide. (v) For $g \in \mathrm{Aut}(\mathbf{G}_0)$,

$$D(\mathbf{G}_R, g^{\mathsf{T}}\mathbf{G}_R g) \le D(\mathbf{G}_R, \mathbf{G}_0) + D(\mathbf{G}_0, g^{\mathsf{T}}\mathbf{G}_0 g) \\ + D(g^{\mathsf{T}}\mathbf{G}_0 g, g^{\mathsf{T}}\mathbf{G}_R g) = 2\nu;$$

for $h \in H_\epsilon^{\pm}(\mathbf{G}_R)$, in the same way, $D(\mathbf{G}_0, h^{\mathsf{T}}\mathbf{G}_0 h) \le \nu + \epsilon + \nu$. The last statement combines the two inclusions. □ The conditions are sufficient, not necessary: the worked cell at $\epsilon = 7.4\%$ has self-maps of price 0, 1.55% and 10.2%, so the condition of (iii) fails, yet $H_\epsilon = \{\mathbf{I}, 2_y\}$ is a group, which the implementation checks directly.

Proposition A11 (Niggli type and the zeros on d). Let (a, b, c) be a Niggli cell, $a\cdot a \le b\cdot b \le c\cdot c$, with Niggli's conditions $|2a\cdot b| \le a\cdot a$, $|2a\cdot c| \le a\cdot a$, $|2b\cdot c| \le b\cdot b$. (i) If it is of type II ($a\cdot b$, $a\cdot c$, $b\cdot c \le 0$), the superbase (d, a, b, c) is obtuse (a known relation between the two reductions; Andrews *et al.*, 2019*a*), and the conorm of d with each vector x is at least $\frac{1}{2}x\cdot x$ minus the smaller of $|x\cdot y|$ and $|x\cdot z|$; so if at most one of $|a\cdot b|$, $|a\cdot c|$, $|b\cdot c|$ exceeds τ and $\tau < \frac{1}{4}a\cdot a$, every conorm on d exceeds every conorm off d that is at most τ. (ii) If it is of type I (all three products positive), none of the four orthogonal superbases $\{\pm a, \pm b, \pm c,\ \mp a \mp b \mp c\}$ is obtuse. (iii) Let $\mathbf{G}^*$ be a lattice with three mutually perpendicular shortest vectors e_1, e_2, e_3, and $\mathbf{G}$ a metric within strain η of it, small enough that every superbase obtuse for $\mathbf{G}$ belongs to the closure of $\mathbf{G}^*$; the largest such η is the strain at which a superbase outside that closure first becomes obtuse near $\mathbf{G}^*$, which the ball enumeration of Lemma A6 computes. If $(e_1\cdot e_2)(e_1\cdot e_3)(e_2\cdot e_3) > 0$ under $\mathbf{G}$, which for the near-orthogonal Niggli cell $(\pm e_1, \pm e_2, \pm e_3)$ is Niggli's type I, the reduced superbase of $\mathbf{G}$, from any starting basis, is a face-diagonal cell, whose three zeros under $\mathbf{G}^*$ form a path through d. *Proof.*

(i) p_{ab}, p_{ac}, $p_{bc} \geq 0$ by the signs; $p_{dx} = x \cdot x + \sum_{y \neq x} x \cdot y \geq x \cdot x - |x \cdot y| - |x \cdot z|$, and the larger of the two is at most $\frac{1}{2} x \cdot x$ by Niggli's conditions and the ordering of the lengths, which gives the bound; if at most one product exceeds τ, the smaller product of x is at most τ, so $p_{dx} \geq \frac{1}{2} x \cdot x - \tau > \tau$ when $\tau < \frac{1}{4} a \cdot a \leq \frac{1}{4} x \cdot x$. (ii) An orthogonal superbase is obtuse only if its three basis products are non-positive, so their product is non-positive; changing the signs of basis vectors leaves $(a \cdot b)(a \cdot c)(b \cdot c)$ unchanged, and it is positive. (iii) A superbase obtuse for $\mathbf{G}$ is in the closure of $\mathbf{G}^*$ by hypothesis. That closure has sixteen superbases, the four orthogonal ones, whose zeros form a triangle, and twelve face-diagonal ones, whose zeros form a path that touches every vector and so d in every labelling; by (ii), applied to the orthogonal triple, only the latter can be obtuse. □

Proposition A12 (the Niggli window; computer-assisted). Fix a zero pattern Z and write lattice vectors in the basis (a, b, c) of a superbase F_0, so that the metrics for which F_0 is obtuse with pattern Z are $\sum_k p_k E_k$, $p_Z = 0$ and $p_k > 0$ off Z, with E_k the metric whose only non-zero conorm is $p_k = 1$. (i) If B and B' are Niggli cells, or more generally Minkowski-reduced bases, of two such metrics, then $B^{-1}B'$ has entries in $\{-1, 0, 1\}$; in particular so has every matrix between two Niggli cells of one lattice. (ii) Let $\mathbf{G}^*$ be such a metric. Let g be the smallest ratio, over the seven parity classes, of the squared length of the shortest vector of the class that is not shortest in it for $\mathbf{G}^*$ to the class minimum, and $\eta_{\mathrm{gap}} = (g - 1)/(g + 1)$; let η_{adm} be the strain from $\mathbf{G}^*$ to the nearest metric for which a basis of vectors shortest in their classes for $\mathbf{G}^*$, Minkowski reduced for no metric with pattern Z, becomes Minkowski reduced; and let $\rho = \min(\eta_{\mathrm{gap}}, \eta_{\mathrm{adm}})$. If two metrics lie within strain ρ of $\mathbf{G}^*$, in a common basis, the matrix between their Niggli cells has entries in $\{-1, 0, 1\}$. *Proof.* (i) A Niggli cell is a Buerger cell, and in three dimensions its vectors realise the successive minima, so it is Minkowski reduced (Gruber, 1973; Křivý & Gruber, 1976). Each vector of a

Minkowski-reduced basis (a, b, c) is shortest in its parity class: a is a shortest vector; a vector of the class of b is not a multiple of a, since multiples of a lie in the classes of 0 and a, so it is at least as long as b; a vector of the class of c is not in the plane of a and b, since that plane meets only the classes of 0, a, b and $a + b$, so it is at least as long as c. By Lemma A1 the vectors shortest in their classes are those whose coefficients relative to F_0 have $\gamma_{ij} = 0$ on every slot off Z, the same for every metric with pattern Z and at most 26 of them. A Minkowski-reduced basis is therefore an ordered basis of these vectors that satisfies the ten conditions $|a| \le |b| \le |c|$, $|b| \le |b \pm a|$ and $|c| \le |c + xa + yb|$ ($x, y \in \{-1, 0, 1\}$, not both zero), which are linear in p, at some p with pattern Z. Over the 38 patterns there are 174 912 ordered bases of such vectors; linear programming retains the 8 256 that satisfy the conditions somewhere, and each of the other 166 656 is excluded only with an exact rational Farkas certificate that the conditions fail at every p with pattern Z, so the retained set contains every Minkowski-reduced basis. All 3 849 216 pairs of retained bases with one pattern differ by a matrix with entries in $\{-1, 0, 1\}$ (Section S11). (ii) By Proposition A4 a strain η multiplies every squared length by a factor in $[1 - \eta, 1 + \eta]$, so below η_{gap} every vector shortest in its class for either metric is shortest in it for $\mathbf{G}^*$; below η_{adm} a basis of such vectors that is Minkowski reduced for either metric is one of the bases retained for Z; and (i) applies to the two Niggli cells, written relative to the obtuse superbase of $\mathbf{G}^*$. □ The radius ρ is computed numerically and to first order in the strain, not certified: 10.4% for the monoclinic lattice of Section 3.2 of the main text, 1.9% for a generic triclinic lattice and 50% for a cubic one (Table S21).

Corollary A13 (from a reduced cell to the Niggli cell; computer-assisted). For every metric and every obtuse superbase F, the Niggli vectors are three of the seven partial sums of F, up to sign. Equivalently, in each of the 24 ordered bases (u_1, u_2, u_3) of F the Niggli cell is $(u_1, u_2, u_3)\,T$, where T is one of the 1 344 unimodular matrices whose

columns are among $\pm u_1$, $\pm u_2$, $\pm u_3$, $\pm(u_1+u_2)$, $\pm(u_1+u_3)$, $\pm(u_2+u_3)$, $\pm(u_1+u_2+u_3)$, written in that basis: one fixed list for every lattice. Since T and $-T$ give the same cell, the 672 with determinant $+1$ suffice. *Proof.* Take for F_0 the reduced superbase of the metric itself and for Z its zero pattern. By the proof of Proposition A12(i) the Niggli cell is one of the bases retained for Z, and F, whose partial sums are shortest in their classes, is one of the 1, 2, 3, 6 or 16 superbases (up to sign) of step (3) of the enumeration. Over the 38 patterns, every obtuse superbase, each of its 24 ordered bases and each retained basis give 6 856 704 matrices; the distinct ones are exactly the 1 344 of the statement (Section S11). □ Testing Niggli's conditions on these 672 bases reads the Niggli cell off the reduced cell; it agreed with the Niggli reduction of Grosse-Kunstleve *et al.* (2004) on every cell tested (Section S11). The list is complete; that it is minimal is not shown. Every member of the closure is thus a landing spot for the Niggli cell. A type II Niggli cell is itself a member; a type I Niggli cell, with three acute angles, is not (as in Proposition A11(ii)), but is one such matrix away.

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